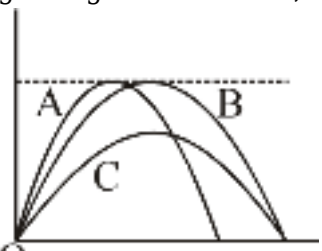
PHYSICS

1) Trajectories are shown in figure for three kicked footballs. Initial vertical & horizontal velocity components are u_y and u_x respectively. Ignoring air resistance, choose the correct statement from List-II for the value of variable in List-I.



	List-I		**List-II**
(A)	time of flight	(I)	greatest for A only
(B)	u_y/u_x	(II)	greatest for C only
(C)	u_x	(III)	equal for A and B
(D)	u_xu_y	(IV)	equal for B and C

(1) (A) → III; (B) → II; (C) → I; (D) → IV

(2) (A) → III; (B) → I; (C) → II; (D) → IV

(3) (A) → I; (B) → III; (C) → II; (D) → IV

(4) (A) → III; (B) → II; (C) → IV; (D) → II

2) If $v^2x^3 = 4$, where v is instantaneous velocity and x is position then instantaneous acceleration of the motion is given by

(1) $a = \frac{-3v^{8/3}}{2^{5/3}}$

(2) $a = \frac{+6}{x^3}$

(3) $a = \frac{-3v^2}{2x^2}$

(4) $a = \frac{v^{5/3}}{2x^2}$

3) A police party is reaching a dacoit in a jeep which is moving at a constant speed v. The dacoit is on a motorcycle. When he is at a distance x from the jeep, he accelerates from rest at a constant rate a. Which of the following relation is true if the police is able to catch the dacoit ?

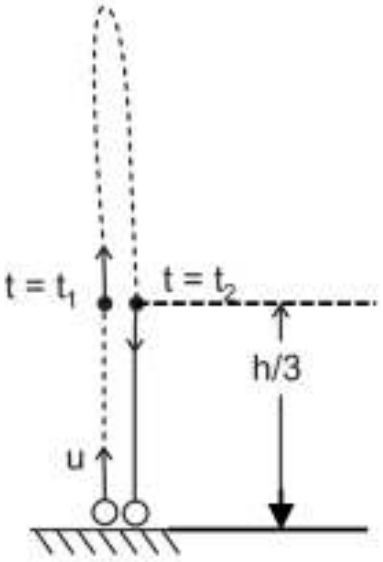
(1) $v^2 \leq ax$

(2) $v^2 \leq 2ax$

(3) $v^2 \geq 2ax$

(4) $v^2 \geq ax$

4) A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h. Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively.



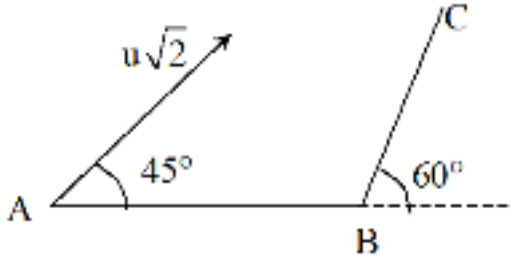
(1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(2) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

(4) $\frac{1}{3}$

5) A particle is projected from a point A with velocity $u\sqrt{2}$ at an angle 45° with horizontal as shown. It strikes the plane BC at right angle. Velocity of the particle at the time of collision is $K\left(\frac{2u}{\sqrt{3}}\right)$ m/s.



Find K.

(1) 1

(2) 2

(3) 3

(4) $\sqrt{2}$

6) A train of length 100 m switches on its headlight when it starts moving with acceleration 0.5 m/s^2 . sometime later, its tail light is switched on. An observer on ground notices that the two events occur at same place. The time interval between two events is.

(1) $10\sqrt{2}$s

(2) 20 s

(3) $20\sqrt{2}$s

(4) 10 s

7) A man is standing on a road and observes that rain is falling at angle 45º with the vertical. The man starts running on the road with constant acceleration 0.5 m/s^2. After a certain time from the start of the motion, it appears to him that rain is still falling at angle 45º with the vertical, with speed $2\sqrt{2}$ m/s . Motion of the man is in the same vertical plane in which the rain is falling. Then which of the following statement(s) are true.
(A) It is not possible
(B) Speed of the rain relative to the ground is 2 m/s.
(C) Speed of the man when he finds rain to be falling at angle 45° with the vertical, is 4m/s.
(D) The man has travelled a distance 16m on the road by the time he again finds rain to be falling at angle 45°.

(1) A, B are true

(2) B, C are true

(3) C, D are true

(4) A, C are true

8) A fighter plane flying horizontally at an altitude of 1.5 km with speed 720 km/h passes directly overhead an anti-aircraft gun. If the gun is fired at an angle of $\sin^{-1}(1/n)$ from the vertical with muzzle speed of shell 600 ms^{-1} to hit the plane. Find n. (Take g = 10 ms^{-2})

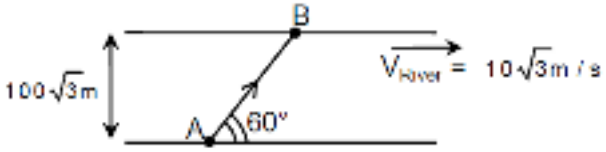
(1) 1

(2) 2

(3) $\sqrt{2}$

(4) 3

9) A swimmer can swim with speed 15 m/sec in still water. He wants to reach at point B as shown in figure. Speed of river is $10\sqrt{3}$ m / s & width of river is $100\sqrt{3}$ m . If time taken by man to cross the river is $\frac{40}{\sqrt{n}}$ sec. Then n is :



(1) 1

(2) 2

(3) 3

(4) 4

10) A projectile is thrown such that its range is 100 m. If it is thrown with the same speed at an angle 5° higher than before, and the range remains same, the first angle of projection was :

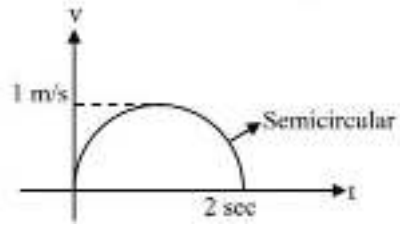
(1) 40°

(2) 42.5°

(3) 45°

(4) 47.5°

11) Which option is correct for the given below -



(1) Particle continuously changes its direction of motion.

(2) Particle reverse its direction of motion once.

(3) Final distance from initial position is $\frac{\pi}{4}$ m

(4) Average speed for first two seconds is $\frac{\pi}{4}$ m/s

12) A particle starts from the origin at t = 0 with an initial velocity of 3.0 $\hat{i}$ m/s and moves in the x-y plane with a constant acceleration $\left(6.0\hat{i}+4.0\hat{j}\right)\text{m/s}^2$. The x-coordinate of the particle at the instant when its y-coordinate is 32 m is D meters. The value of D is :-

(1) 50

(2) 32

(3) 60

(4) 40

13) A particle moves in the *x*-*y* plane. It's *x* and *y* coordinates vary with time *t* according to equations as $x = t^2 + 2t$ and $y = 2t$. Possible shape of path followed by the particle is

(1) Straight line

(2) Circle

(3) Parabola

(4) More information is required to decide

14) A body is thrown vertically up with a velocity u. It passes three points A, B and C in its upward journey with velocities $\frac{u}{2}, \frac{u}{3}$ and $\frac{u}{4}$ respectively. The ratio of the separations between points A and B and between A and C, i.e., $\frac{AB}{AC}$ is :-

(1) 1

(2) 2

(3) $\frac{10}{7}$

(4) $\frac{20}{27}$

15) There are four vernier scales. Specifications of instrument is given in List-I & least count is given in List-II [S = main scale division, n = number of vernier scale divisions coinciding with (n – 1) main scale divisions]

	List-I		**List-II**
(P)	S = 1mm ; n = 10	(A)	0.05 mm
(Q)	S = 0.5 mm ; n = 10	(B)	0.01 mm
(R)	S = 0.5mm; n = 20	(C)	0.1 mm
(S)	S = 1mm ; n = 100	(D)	0.025 mm

(1) P→A; Q→B; R→C; S→D

(2) P→C; Q→A; R→D; S→B

(3) P→C; Q→A; R→B; S→D

(4) P→C; Q→B; R→D; S→A

16) The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a percentage error of 2% then percentage error in density is :-

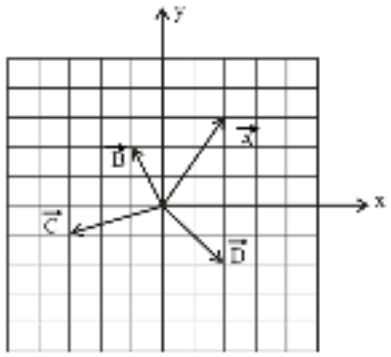
(1) 0.9 %

(2) 2.4 %

(3) 3.11 %

(4) 4.2 %

17) Four vectors are as shown in the Cartesian coordinate system. All the vectors are drawn on scale.



The direction of $\vec{A}+\vec{B}+\vec{C}+\vec{D}$ will be :-

(1) →

(2) ↑

(3) ←

(4) ↓

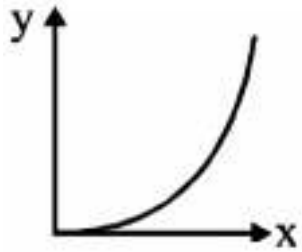
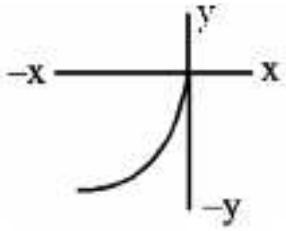
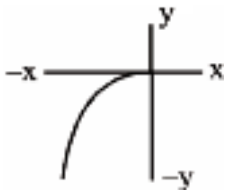
18) Consider three vectors $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = \hat{i} - \hat{j} - \hat{k}$ and $\vec{C} = 2\hat{i} - 3\hat{j} - 4\hat{k}$. If $\alpha\vec{A} + \beta\vec{B}$ is perpendicular to $\vec{C}$, α & β are numbers, the ratio of α and β.

(1) 7 : 9
(2) – 7 : 9
(3) 9 :7
(4) – 9 : 7

19) Given that $y = \sin 3x + \frac{4}{3}\cos 3x$. What is the maximum rate of change in y with respect to x ?

(1) 7
(2) 4
(3) 5
(4) 6

20) Which of the following match is correct ?

(i)		**(a)**	$y = \frac{x^2}{2}$
(ii)		**(b)**	$y = -2x^2$
(iii)		**(c)**	$2x = y^2$
(iv)		**(d)**	$3x = -2y^2$

(1) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)
(2) (i)-(a), (ii)-(c), (iii)-(b), (iv)-(d)
(3) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)
(4) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b)

21) $y = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x} + 1\right)$. Find $\frac{dy}{dx}$

(1) $\frac{dy}{dx} = 1 + x + \frac{2}{x^3} - \frac{1}{x^2}$

(2) $\frac{dy}{dx} = 1 + 2x + \frac{2}{x^3} - \frac{1}{x^2}$

(3) $\frac{dy}{dx} = 1 + 2x + \frac{2}{x^3} + \frac{1}{x^2}$

(4) $\frac{dy}{dx} = 1 + 2x - \frac{2}{x^3} - \frac{1}{x^2}$

22) The smallest division on the main scale of a vernier callipers is 1 mm, and 10 vernier divisions coincide with 9 main scale divisions. While measuring the diameter of a sphere, the zero mark of the vernier scale lies between 2.0 and 2.1 cm and the fifth division of the vernier scale coincide with a scale division. Then diameter of the sphere is

(1) 2.05 cm

(2) 3.05 cm

(3) 2.50 cm

(4) None of these

23) A student determined Young's modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of g is taken to be 9.8 m/s^2 without any significant error, his observations are as follows. Then the fractional error in the measurement of Y is :

Physical Quantity	**Least count of the Equipment used for measurement**	**Observed value**
Mass (M)	1 g	2 kg
Length of bar (L)	1 mm	1 m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

(1) 0.0083

(2) 0.0155

(3) 0.155

(4) 0.083

24) The relative density of a material is found by weighing the body first in air and then in water. If weight in air is (15.0 ± 0.3) N and weight in water is (10.0 ± 0.2) N then the maximum permissible percentage error in relative density is:

(1) 2%

(2) 4%

(3) 10%

(4) 12%

25) The pitch of a screw gauge having 50 divisions on its circular scale is 1 mm. When the two jaws of the screw gauge are in contact with each other, the zero of the circular scale lies 6 division below the line of graduation and zero of linear scale is visible. When a wire is placed between the jaws, 3 linear scale divisions are clearly visible while 31st division on the circular scale coincide with the reference line. The diameter of the wire is :

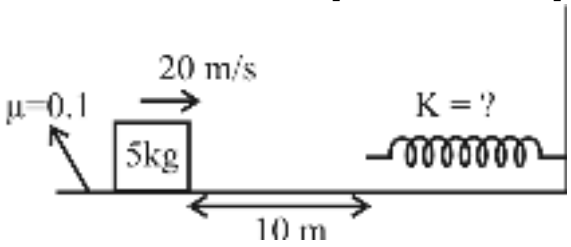
(1) 3.62 mm

(2) 3.50 mm

(3) 3.72 mm

(4) 3.74 mm

26) If the maximum compression in the spring is 50 cm then find the value of K($g = 10\ m/s^2$) :-



(1) $7580\frac{N}{m}$

(2) $7600\frac{N}{m}$

(3) $8420\frac{N}{m}$

(4) None of these

27) The potential energy (in joules) function of a particle in a region of space is given as :
$U = (2x^2 + 3y^3 + 2z)$
Here x, y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P (1m, 2m, 3m).

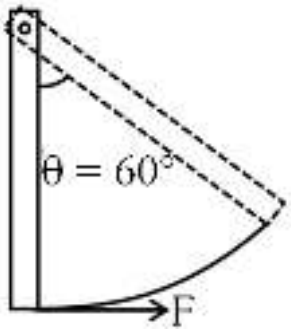
(1) 2

(2) 3

(3) 0

(4) 4

28) Find the work done by applied force F by slowly bringing the rod to the inclined position. mass of rod is 5 kg & length 1 m. ($g = 10\ m/sec^2$) :-



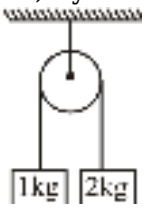
(1) 205 J

(2) 5 J

(3) 10 J

(4) 12.5 J

29) System is released from rest at t = 0. Friction is absent and string is massless. In time t = 0.3 s



(a)	Work done by gravity on 2 kg block	(p)	–1.5 J
(b)	Work done by gravity on 1 kg block	(q)	2 J
(c)	Work done by string on 2 kg block	(r)	3 J
(d)	Work done by string on 1 kg block	(s)	–2 J

(1) a→r, b→p, c→s, d→q

(2) a→p, b→r, c→s, d→q

(3) a→r, b→p, c→q, d→s

(4) a→r, b→q, c→p, d→s

30) A 50 kg girl is swinging on a swing from rest. Then the power delivered when she was moving with a velocity of 2m/s upwards in a direction making an angle 60º with the vertical is :-

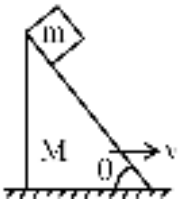
(1) 980 W

(2) 490 W

(3) 480 W

(4) 245 W

31) A block of mass m is stationary with respect to a wedge of mass M moving with uniform speed v on a horizontal surface. Find the work done by friction force on the block in t second



(1) zero

(2) $-\frac{mgvt}{2}\sin 2\theta$

(3) $-\frac{mgvt}{2}$

(4) $-\frac{mgvt}{2}\sin\theta$

32) ***Assertion (A) :-*** Work done by spring force may be positive.

Reason (R) :- When spring is shaping in its natural length from an extension or compression, then work done by spring force is positive.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

33) Water falls from a height of 100 m at the rate of 10 kg/s to operate a turbine. The losses due to fricitional force are 20% of the input energy. How much power is generated by the turbine? (g = 10 m/s^2)

(1) 10 kW

(2) 8 kW

(3) 12 kW

(4) 7 kW

34) A balloon and its content having mass M is moving up with an acceleration 'a'. The mass that must be released from the content so that the balloon starts moving up with an acceleration '3a' will be (Take ' g ' as acceleration due to gravity) :-

(1) $\frac{2Ma}{3a+g}$

(2) $\frac{3Ma}{2a-g}$

(3) $\frac{3Ma}{2a+g}$

(4) $\frac{2Ma}{3a-g}$

35) A pen of mass m is lying on a piece of paper of mass M placed on a rough table. If the coefficients of friction between the pen and paper and the paper and the table are μ_1 and μ_2, respectively. Then, the minimum horizontal force with which the paper has to be pulled for the pen to start slipping is given by :

(1) $(m + M)(\mu_1 + \mu_2) g$

(2) $(m\mu_1 + M\mu_2) g$

(3) $(m\mu_1 + (m + M)\mu_2) g$

(4) $m(\mu_1 - \mu_2) g$

36) A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is :

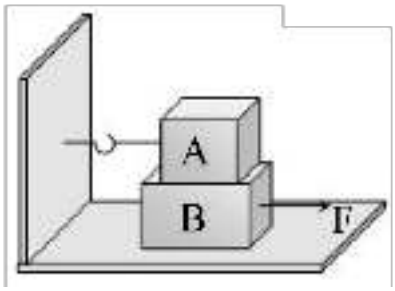
(1) $\frac{2}{3}$ m

(2) $\frac{1}{3}$ m

(3) $\frac{1}{2}$ m

(4) $\frac{1}{6}$ m

37) A block A with mass 100 kg is resting on another block B of mass 200 kg. As shown in figure a horizontal rope tied to a wall holds it. The coefficient of friction between A and B is 0.2 while coefficient of friction between B and the ground is 0.3. The minimum required force F to start moving B will be ..... N :-



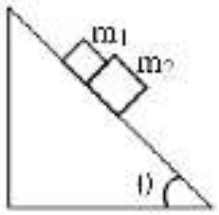
(1) 900

(2) 100

(3) 1100

(4) 1200

38) A block of mass m_1 = 1 kg another mass m_2 = 2 kg, are placed together (see figure) on an inclined plane with angle of inclination θ. Various values of θ are given in List-I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to μ = 0.3. In List-II expression for the friction on block m_2 given. Match the correct expression of the friction in List-II with the angles given in List-I, and choose the correct option.
The acceleration due to gravity is denoted by g.
[Useful information : tan (5.5°) ≈ 0.1 ; tan (11.5°) ≈ 0.2 ; tan (16.5° ≈ 0.3)]



	List-I		List-II
P.	θ = 5°	A.	$m_2 g\sin\theta$
Q.	θ = 10°	B.	$(m_1 + m_2)\, g\sin\theta$
R.	θ = 15°	C.	$\mu m_2 g\cos\theta$
S.	θ = 20°	D.	$\mu(m_1 + m_2)g\cos\theta$

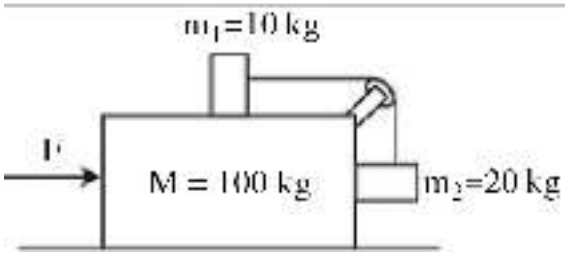
(1) P-A, Q-A, R-A, S-C

(2) P-B, Q-B, R-B, S-C

(3) P-B, Q-B, R-B, S-D

(4) P-B, Q-B, R-C, S-C

39) Three masses M = 100 kg, m_1 = 10 kg and m_2 = 20 kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2}. The value of F is ..... N :-



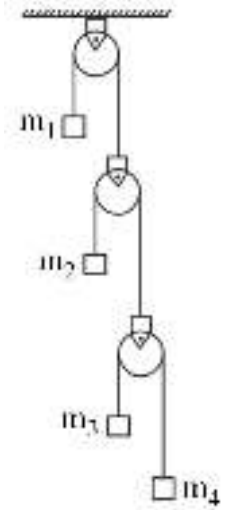
(1) 3360

(2) 3380

(3) 3120

(4) 3240

40) In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement ?



(1) $4a_1 + 2a_2 + a_3 + a_4 = 0$

(2) $a_1 + 4a_2 + 3a_3 + a_4 = 0$

(3) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

(4) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

41) A trolley of mass 300 kg carrying a sandbag of 25 kg is moving uniformly with a speed of 27 km/h on a frictionless track. After a while, sand starts leaking out of a hole on the floor of the trolley at the rate of 0.05 kgs^{-1}. What is the speed (in km/h) of the trolley after the entire sand bag is empty ?

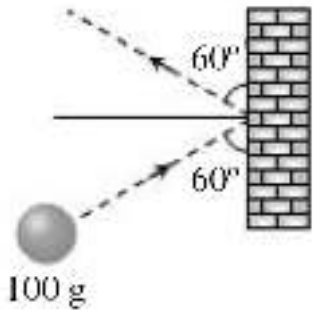
(1) 36

(2) 9

(3) 54

(4) 27

42) A mass of 100 g strikes the wall with speed 5 m/s at an angle as shown in figure and it rebounds with the same speed. If the contact time is 2×10^{-3} sec, what is the force applied on the mass by the wall :-



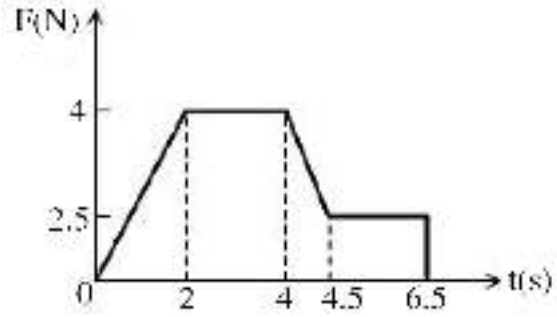
(1) $250\sqrt{3}$ N to right

(2) 250 N to right

(3) $250\sqrt{3}$ N to left

(4) 250 N to left

43) A body of 2 kg has an initial speed 5 ms^{-1}. A force acts on it for some time in the direction of motion. The force time graph is shown in figure. The final speed of the body. ..... ms^{-1} :-



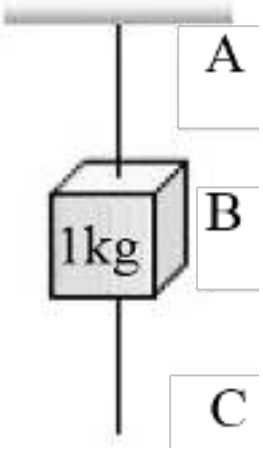
(1) 9.25

(2) 5

(3) 14.3

(4) 4.25

44) A mass of 1 kg is suspended by a string A. Another string C is connected to its lower end (see figure)., if the string C is stretched slowly, then :-



(1) The portion AB of the string will break

(2) The portion BC of the string will break

(3) None of the strings will break

(4) None of the above

45) A bird weighs 2 kg and is inside a closed cage of 1 kg. If it starts flying upwards with constant velocity 1 m/s, then what is the weight of the bird and cage assembly ..... kg :-

(1) 1.5

(2) 2.5

(3) 3

(4) 4

CHEMISTRY

1) From separate solutions of four sodium salts NaW, NaX, NaY and NaZ had pH 7.0, 9.0, 10.0 and 11.0 respectively, when each solution was 0.1 M, the weakest acid is :

(1) HW

(2) HX

(3) HY

(4) HZ

2) How many grams of NaCN are present in 500 mL of its aqueous solution in which it is 0.01% hydrolysed at 25°C ? (Given pK_a for HCN = 6)

(1) 49 g

(2) 24.5 g

(3) 4.9 g

(4) 2.45 g

3) Calculate the pH of a buffer solution prepared by dissolving equal volumes of 4M NaCN in an aqueous solution containing 2M HCl.
$(K_a(HCN) = 4 \times 10^{-6})$:-

(1) 6.4

(2) 5.4

(3) 8.4

(4) 12.3

4) Which of the following mixtures of solutions can function as a buffer solution ?

(1) 50 ml of 0.2 M NH_3 + 50 ml of 0.1 M HCl

(2) 50 ml of 0.2 M NH_3 + 50 ml of 0.2 M HCl

(3) 50 ml of 0.2 M HCl + 05 ml of 0.2 M NaOH

(4) 50 ml of 0.1 CH_3COOH + 50 ml of 0.1 M HCl

5) Match List I with List II and choose the correct answer from the code :

	List I Electrolyte		**List II Solubility product**
(a)	Bi_2S_3	(i)	$4s^3$
(b)	CdS	(ii)	$27s^4$
(c)	$Al(OH)_3$	(iii)	$108s^5$
(d)	CaF_2	(iv)	s^2

Code : (a) (b) (c) (d)
(1) (iii) (iv) (ii) (i)
(2) (i) (ii) (iii) (iv)
(3) (iv) (i) (ii) (iii) (4) (ii) (iii) (i) (iv)

(1) 1
(2) 2
(3) 3
(4) 4

6) What will happen at 25°C if pH of solution of 0.01 M $Mg(NO_3)_2$ solution is adjusted to pH = 11 :-
(K_{sp} Mg $(OH)_2 = 1 \times 10^{-11}$)

(1) Solution will be saturated
(2) ppt will take place
(3) ppt will not take place
(4) None of these

7) At 25°C 200 ml of an aqueous solution of HCl (pH=2) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12). Thus, pH of the resultant mixture is :-

(1) 14.0
(2) 10.2
(3) 7.9
(4) 11.3

8) 20 ml, 0.1M HA [$(K_b)_{A^-} = 10^{-10}$] is titrated
against 0.2M, NaOH. Select the correct statement :-
[Given : log2 = 0.3, log3 = 0.48]

(1) Initially pH of HA solution is 11.5
(2) At half equivalence point pH of solution is 4 and at equivalence point pH is 9.09
(3) Phenolphthalein is suitable indicator for this titration but methyl orange is not suitable
(4) Phenolphthalein & methyl orange both are suitable indicator for end point determination

9) K_b for NH_4OH is 1.8×10^{-5}, what is the $[OH^-]$ in its solution of 0.1 M concentration :-

(1) 1.8×10^{-6} M

(2) 1.34×10^{-3} M

(3) 5×10^{-4} M

(4) 4.2×10^{-3} M

10) 50 ml of 0.2 M HCl is mixed with 50 ml of 0.2 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$). Thus, pH of the resulting solution is :-

(1) 7.00

(2) 1.7

(3) 0.30

(4) 1.00

11) ***Assertion (A) :-*** When small amount of strong acid or strong base is added to pure water, it's pH undergoes a change.
Reason (R) :- Addition of an acid or base increases the degree of ionisation of water.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

12) Reaction $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ degree of dissociation (α) is terms of K_p and total equilibrium pressure P is :-

(1) $\alpha = \sqrt{\frac{4P + K_P}{K_P}}$

(2) $\alpha = \sqrt{\frac{K_P}{4P + K_P}}$

(3) $\alpha = \sqrt{\frac{K_P}{4P}}$

(4) None

13) 1.1 mole of A are mixed with 2.2 moles of B and the mixture is kept in a 1 L vessel till the equilibrium is established in the reaction. If the molar conc. of C at the equilibrium point is 0.2 M then the value of equilibrium constant for the gaseous reaction $A + 2B \rightleftharpoons 2C + D$

(1) 0.001

(2) 0.002

(3) 0.003

(4) 0.004

14) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

Initially the molar ratio of N_2 and H_2 was 1 : 3. At equilibrium 50% of each reactant has reacted. If the equilibrium pressure is P then the partial pressure of NH_3 at equilibrium is :-

(1) $\frac{P}{3}$

(2) $\frac{P}{4}$

(3) $\frac{P}{6}$

(4) $\frac{P}{8}$

15) What will be the direction of reaction if concentration of H_2, I_2 and HI are 2M, 2M and 8M respectively. K_C for reaction $H_2 + I_2 \rightleftharpoons 2HI$ is 4.

(1) forward direction

(2) backward direction

(3) equilibrium condition

(4) reaction will be completed

16) For an equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$, which of the following statement is true ?

(1) The pressure changes do not affect the equilibrium

(2) More of ice melts if pressure on the system is increased

(3) More of liquid freezes if pressure on the system is increased

(4) The pressure increase may increase or decrease the degree of advancement of the process

17) Addition of which substance at equilibrium will decrease the intensity of red colour in the following reaction :

$$\underset{\text{Yellow}}{Fe^{+3}_{(aq)}} + \underset{\text{Colour less}}{SCN^{\ominus}_{(aq)}} \rightleftharpoons \underset{\text{Deep red}}{[Fe(SCN)]^{-2}_{(aq)}}$$

(1) Oxalic acid

(2) KSCN

(3) $HgCl_2$

(4) Both (1) & (3)

18) For the reaction,
$2Fe(NO_3)_3 + 3Na_2CO_3 \rightarrow Fe_2(CO_3)_3 + 6NaNO_3$
initially 3 mole $Fe(NO_3)_3$ and 4 mole of Na_2CO_3 are taken. If 6.4 mole of $NaNO_3$ is obtained then % yield of given reaction is

(1) 50

(2) 84

(3) 80

(4) 100

19) 20.0 g of a magnesium carbonate sample decomposes on heating to given carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample ?
$MgCO_3 \rightarrow MgO + CO_2$

(1) 84

(2) 75

(3) 96

(4) 60

20) Match List-I with List-II for the reaction
$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(g)}$

	List-I		**List-II**
(a)	8 moles H_2O produced from ____ $C_3H_8(g)$ at 1 atm and 0°C	(i)	72g
(b)	88g $C_3H_8(g)$ reacts with ____ O_2	(ii)	3 mole
(c)	44g C_3H_8 produces ____ CO_2	(iii)	44.8L
(d)	1 mole C_3H_8 on reaction with 10 moles O_2 produces _____ H_2O	(iv)	320 g

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

21) A compound contains 54.55% carbon, 9.09% hydrogen and 36.36% oxygen. The empirical formula of this compound is :

(1) C_3H_5O

(2) $C_4H_8O_2$

(3) $C_2H_4O_2$

(4) C_2H_4O

22) If the atomic mass unit (u) were defined to be one fifth of the mass of C-12, what would be the atomic weight of N in amu on new scale :-

(1) 33.6

(2) 5.83

(3) 11.67

(4) 2.80

23) How many moles of H_2SO_3 is required to reduce one mol of $KMnO_4$ in acidic medium :-

(1) 1.5 mol

(2) 3 mol

(3) 2.5 mol

(4) 1 mol

24) ClO_3^- + I_2 + $xH_2O \rightarrow IO_3^-$ + Cl^- + yH^+ The values of x and y in balanced redox reaction will be respectively:-

(1) 4,2

(2) 1,3

(3) 3,6

(4) None of these

25) The oxidation states of S-atoms in caro's and marshal's acid are :-

(1) + 6, +6

(2) +6, +4

(3) +6, –6

(4) +4, +6

26)

Which of the following is a Redox reaction:

(1) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(2) $2CuO(s) + Cu_2S(s) \rightarrow 4Cu(s)+SO_2(g)$

(3) $PbO + 2HCl \rightarrow PbCl_2 + H_2O$

(4) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

27) Calculate the enthalpy change when 50 mL of 0.01 M $Ba(OH)_2$ reacts with 25 mL of 0.01 M HCl. Given that ΔH^0neutralisation of strong acid and strong base is 140 kcal eq^{-1} :

(1) 14 kcal

(2) 35 cal

(3) 10 cal

(4) 7.5 cal

28) If enthalpies of formation of $C_2H_{4(g)}$, $CO_{2(g)}$ and $H_2O_{(\square)}$ at 25°C and 1 atm pressure be 52, –394 and –286 kJ mol^{-1} respectively, the enthalpy of combustion of $C_2H_{4(g)}$ will be :-

(1) +1412 kJ mol^{-1}

(2) –1412 kJ mol^{-1}

(3) +141.2 kJ mol^{-1}

(4) –141.2 kJ mol^{-1}

29) $XeF_{2(g)} + H_{2(g)} \rightarrow 2HF_{(g)} + Xe_{(g)}$

$\Delta H° = –430$ kJ

Bond energy → H-H = 435 kJ/mol

→ H-F = 565 kJ/mol

Calculate average bond energy of Xe-F bond:-

(1) 267 kJ/mol

(2) 562.5 kJ/mol

(3) 132.5 kJ/mol

(4) 267.3 kJ/mol

30) What amount of energy (kJ) is released in the combustion of 5.8 g of $C_4H_{10}(g)$?

$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$; $\Delta H° = -5756$ kJ

(1) 575.6

(2) 287.8

(3) 182

(4) 57.56

31) Based on following thermo chemical reaction :-

$H_2O(g) + C(s) \rightarrow CO(g) + H_2(g)$; $\Delta H = 131$ KJ

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$; $\Delta H = -282$ KJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H = -242$ KJ

$C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = x$ KJ

The value of x will be

(1) –393 KJ

(2) –655 KJ

(3) +393 KJ

(4) +655 KJ

32) If bond enthalpies of N ≡ N, H – H and N – H bonds are x_1, x_2 and x_3 respectively, $\Delta H°_f$ for $NH_{3(g)}$ will be

(1) $x_1 + 3x_2 - 6x_3$

(2) $\frac{x_1}{2} + \frac{3x_2}{2} - 3x_3$

(3) $3x_3 - \frac{x_1}{2} - \frac{3x_2}{2}$

(4) $6x_3 - x_1 - 3x_2$

33) The heat of solution of $Na_2SO_4(s)$ and $Na_2SO_4.10H_2O(s)$ are –2.34 kJ mol^{-1} and 78.87 kJ mol^{-1} respectively. The heat of hydration of $Na_2SO_4(s)$ is :-

(1) –81.21 kJ mol^{-1}

(2) 75.42 kJ mol^{-1}

(3) 81.21 kJ mol^{-1}

(4) –75.82 kJ mol^{-1}

34) $Na(s) \rightarrow Na(g)$, the heat of reaction is called as

(1) Heat of vaporisation

(2) Heat of atomisation

(3) Heat of sublimation

(4) Both (2) and (3)

35)

List-I		List-II	
(P)	Adiabatic expansion of an ideal gas against vaccum	(1)	$\Delta S_{sys} > 0$
(Q)	Melting of ice at 1 atm & 0°C	(2)	$\Delta S_{SYS} = 0$
(R)	Adiabatic reversible compression involving an ideal gas	(3)	$\Delta S_{uni} = 0$
(S)	Isothermal reversible expansion of an ideal gas	(4)	$\Delta S_{surr} < 0$

(1) P-1; Q-3; R-2; S-4

(2) P-1; Q-2; R-4; S-1

(3) P-2; Q-3; R-1; S-4

(4) P-3; Q-2; R-4; S-1

36) Which one of the following statement is **INCORRECT** ?

(1) The net increase in entropy of a system is zero in any reversible Cyclic process.

(2) At constant temperature & pressure decrease in gibbs free energy represent useful work done by system.

(3) The change in Gibb's free energy with pressure for one mole ideal gas at constant temperature is $\Delta G = RT\ell n\frac{P_2}{P_1}$

(4) For a spontaneous change $(\Delta G)_{T,P} > 0$

37) Heat of combustion of ethanol is –671.08 kcal mol^{-1} at 27°C. What is ΔE at 27°C for this reaction.

(1) –335.24 kcal mol^{-1}

(2) –670.48 kcal mol^{-1}

(3) +335.24 kcal mol^{-1}

(4) +670.48 kcal mol^{-1}

38) 3 moles of an ideal gas at 200 K is compressed reversibly and adiabatically untill final temperature reaches 250 K. If molar heat capacity at constant volume is 27 cal/mol.K find work.

(1) 3050 cal

(2) 3550 cal

(3) 4050 cal

(4) 4550 cal

39)

One litre of an ideal gas at pressure of 5 atm and temperature of 27°C expands into a vaccum until its total volume becomes 5 litres. Which of the following is not correct for this process if this process is isothermal.

(1) Work is zero

(2) Heat transfer is zero

(3) Change in internal energy is zero

(4) Entropy change is zero

40) Two moles ideal gas is expanded isobaric and reversibly from 200K to 300K. Work done by the gas would be :

(1) 400 cal

(2) – 400 cal

(3) 1660 cal

(4) –1660 cal

41) For the given reaction :
$H_2(g) + S(s) \rightarrow H_2S(g)$; $\Delta H_r = 100$ kJ/mol and $\Delta S_r = 400$ J/mol/K
Temperature at which above reaction occurs spontaneously is (Assuming ΔH_r and ΔS_r are independent of temperature)

(1) 250 K

(2) 260 K

(3) 200 K

(4) 240 K

42) If $\Delta H_{system} = 2000 \frac{KJ}{mol}$. Then calculate ΔS_{surr} at 400 K :-
(assume system - surrounding is at equilibrium)

(1) $-35 \frac{KJ}{mol\,K}$

(2) $-5 \frac{KJ}{mol\,K}$

(3) $-16000 \frac{KJ}{mol\,K}$

(4) $-20 \frac{kJ}{mol\,K}$

43) When zinc reacts with HCl which produce H_2 gas due to which system got expanded by 10 L against 1 atm. If the change in internal energy is –2 kJ then what would be the change in enthalpy ?
(1 L atm = 100 J)

(1) –3 kJ

(2) –1 kJ

(3) 1 kJ

(4) 3 kJ

44)

In the graph between ΔG° (in cal) and T (in kalvin) slope is ; –1 what will value of ΔG° at 300 K Temperature ? if ΔH° = – 500 cal/mol

(1) –100 cal/mol

(2) –800 cal/mol

(3) –400 cal/mol

(4) –300 cal/mol

45) **Assertion :** If volume is kept constant and an inert gas such as argon is added which does not take part in the reaction, the equilibrium remains undisturbed.
Reason : Addition of an inert gas at constant volume does not change the partial pressure or the molar concentrations of the substance involved in the reaction.

(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.

(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion

(3) Assertion is correct, reason is incorrect

(4) Assertion is incorrect, reason is correct.

BIOLOGY

1) Among the following, select the correct statements.
A. In majority of higher plants & animals, growth & reproduction are mutually exclusive event.
B. In non living objects growth is by accumulation of material on the surface.
C. An isolated metabolic reaction outside the body of an organism performed in a test tube neither living nor non-living.
D. All organisms from prokaryotes to eukaryotes can sense and respond to environmental cues.

(1) B, C, D only

(2) A, B, C only

(3) A, D, C only

(4) A, B, C, D

2) **Statement-I :** Artificial classification is based on a few morphological characters.
Statement-II : Natural classification is based on overall similarities or gross morphology among the organism.

(1) Statement I & II both are correct

(2) Statement I & II both are incorrect

(3) Statement I is correct & II is incorrect

(4) Statement II is correct & I is incorrect

3) Which of the following statement is incorrect ?

(1) Biological names are generally in English and written in Italics.

(2) Biological names are printed in Italics to indicate the importance of Latin language.

(3) Biological names when hand written then generic name and specific epithet are separately underlined.

(4) Biological names are derived from Latin or are Latinized

4) Match the column correctly :

	Column I (Pathogen)		**Column II (Disease)**
(A)	*Albugo*	(i)	Late Blight of potato
(B)	*Puccinia*	(ii)	White Rust of crucifers
(C)	*Alternaria*	(iii)	Early Blight of Potato
(D)	*Phytophthora*	(iv)	Rust disease in wheat

(1) A-(iv), B-(iii), C-(ii), D-(i)

(2) A-(ii), B-(iv), C-(iii), D-(i)

(3) A-(iii), B-(ii), C-(i), D-(iv)

(4) A-(ii), B-(i), C-(iv), D-(iii)

5) Choose the incorrect statement from following w.r.t. bacteria-

(1) Ribosomes are the site of protein synthesis

(2) Pili do not play a role in motility

(3) Inclusion bodies are bounded by membrane

(4) Cell wall prevent the bacterium from bursting and collapsing

6) Read the following statements and choose the correct option-
Statement-I : Antibiotic resistance is a unique character conferred by plasmid to bacteria.
Statement-II : Plasmid represent the genomic DNA of bacteria.

(1) Only statement-I is correct.

(2) Only statement-II is correct.

(3) Both statement I and II are correct.

(4) None of the above statement is correct

7) Given below are two statements :-
Statement-I : According to five kingdom classification *chlamydomonas* and *chlorella* are placed in kingdom protista.
Statement-II : In five kingdom classification virus, viroids, prions and lichens are placed in kingdom monera.
In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I and statement-II are incorrect.

(4) Statement-I is correct but statement-II is incorrect.

8) Match the column I with column II :-

Column-I		**Column-II**	
(i)	Dianoflagellates	(a)	Pellicle
(ii)	Diatoms	(b)	Saprotrophic
(iii)	Slime moulds	(c)	Cellulosic cell wall
(iv)	Euglenoids	(d)	Cellulosic cell wall with Silica

(1) i-d, ii-c, iii-b, iv-a

(2) i-c, ii-d, iii-b, iv-a

(3) i-c, ii-d, iii-a, iv-b

(4) i-d, ii-c, iii-a, iv-b

9)

Study the following statements :-
(A) Lichens are symbiotic association between algae and fungi
(B) Algal component of lichen is known as mycobiont and fungal component is known as phycobiont.
(C) Lichens are used as pollution indicators
(D) Mycoplasma can survive without oxygen.

How many of the above statement(s) is/are correct ?

(1) One

(2) Two

(3) Three

(4) Four

10) Match the column(I) with column(II) and select the correct option given below :-

Column-I		**Column-II**	
A.	T.O. Diener	P.	Term virus
B.	W.M. Stanley	Q.	Contagium vivium fluidum
C.	D.J. Iwanowsky	R.	Crystalization of virus
D.	M.W. Beijerinek	S.	Discovery of viroid
		T.	Discovery of virus

Option :

(1) A-S, B-R, C-P, D-Q

(2) A-T, B-R, C-P, D-Q

(3) A-S, B-Q, C-T, D-P

(4) A-S, B-R, C-T, D-Q

11) Consider the following four (A-D) statements.
(A) Spores of slime moulds do not have any walls
(B) *Euglena* shows autotrophic as well as heterotrophic nutrition
(C) Dinoflagellates have plated cell wall in which silica particles are embedded
(D) Diatoms lack flagella/cilia but they float passively on water surface
Which of the above statement(s) is/are **not** correct?

(1) A and C
(2) A, C and D
(3) B and C
(4) B and D

12)

Which statement is/are wrong with respect to Archaebacteria?
(a) Archaebacteria differ from other bacteria in having a different cell wall structure
(b) Feature of cell wall is responsible for their survival in extreme condition
(c) Methanogens are present in the guts of ruminant animal such as cow & buffalo
(d) They are responsible for the production of methane.

(e) Archaebacteria are sole members of the kingdom monera.
(1) a and e
(2) b and e
(3) a, b and c
(4) only e

13)

Match the following columns.

Column-I		**Column-II**	
A	Imperfect fungi	1	*Neurospora*
B	Pathogen of smut disease	2	*Ustilago*
C	Used in genetic work	3	Morels and truffles
D	Edible delicacies	4	*Trichoderma*
E	Bread mould	5	*Rhizopus*

(1) A= 3 B= 1 C= 4 D= 2 E= 5
(2) A= 1 B= 3 C= 5 D= 4 E= 2
(3) A= 2 B= 1 C= 3 D= 4 E= 5
(4) A= 4 B= 2 C= 1 D= 3 E= 5

14) Match the following columns :-

Column-A		Column-B	
(1)	*Sequoia*	(A)	Coralloid root
(2)	*Cycas*	(B)	Mycorrhiza
(3)	*Pinus*	(C)	Tallest tree
(4)	Mango	(D)	Embryo sac

(1) 1-C, 2-A, 3-B, 4-D

(2) 1-B, 2-C, 3-A, 4-D

(3) 1-A, 2-B, 3-C, 4-D

(4) 1-D, 2-C, 3-A, 4-B

15) Find the incorrect match:

	Group	**Flagellar number and position of insertions**
(1)	Rhodophyceae	2-8, Equal, Apical
(2)	Chlorophyceae	2-8, Equal, Apical
(3)	Phaeophyceae	2, Unequal, Lateral
(4)	Dinoflagellates	1 longitudinal, 1 transvers

(1) 1

(2) 2

(3) 3

(4) 4

16) Match the column A with column B.

A		B	
P	*Ectocarpus*	(i)	Green algae
Q	*Spirogyra*	(ii)	Brown algae
R	*Porphyra*	(iii)	Red algae
S	*Ficus*	(iv)	Angiosperm

Choose the correct answer from the options given bellow –

	(P)	**(Q)**	**(R)**	**(S)**
(1)	(ii)	(i)	(iii)	(iv)
(2)	(i)	(ii)	(iii)	(iv)

(3)	(ii)	(iii)	(i)	(iv)
(4)	(iii)	(ii)	(i)	(iv)

(1) 1

(2) 2

(3) 3

(4) 4

17) Choose the incorrect match:-

(1) Isogamy (flagellated Gametes) - *Ulothrix*

(2) Isogamy (Non-flagellated gametes) - *Spirogyra*

(3) Anisogamy - *Sargassum*

(4) Oogamy - *Fucus*

18) Match List-A, B and C :

	List-A		**List-B**		**List-C**
A	Chlorophyceae	P	Chlorophyll a, d	I	Cyanophycean starch
B	Cyanobacteria	Q	Chlorophyll a, b	II	Starch
C	Phaeophyceae	R	Chlorophyll a, c	III	Mannitol
D	Rhodophyceae	S	C-Phycocyanin	IV	Floridean starch

(1) A-Q-II, B-R-III, C-S-I, D-P-IV

(2) A-S-I, B-R-III, C-P-IV, D-Q-II

(3) A-S-I, B-Q-II, C-P-IV, D-R-III

(4) A-Q-II, B-S-I, C-R-III, D-P-IV

19) **Assertion (A) :** In Gymnosperm, root in some genera have fungal association in the form of mycorrhiza.
Reason (R) : In Gymnosperm generally tap root present.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

20) Consider the following w.r.t pteridophytes
Statement I: The diploid sporophyte is represented by dominant, independent, photosynthetic, vascular plant body.
Statement II: It alternates with multicellular, autotrophic, independent but short-lived haploid gametophyte.

(1) Both are correct

(2) Both are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

21) **Assertion :** In gymnosperms male and female gametophyte do not have independent free living existence.
Reason : They remain within sporangia retained on the sporophyte.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

22) **Assertion :** Mosses reduce the impact of falling rain and prevent soil erosion.
Reason : Mosses form dense mats on the soil.

(1) Both assertion & reason are true & the reason is correct explanation of assertion.

(2) Both assertion & reason are true but reason is not correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Both assertion & reason are false.

23) Given below are two statements : one is labelled as assertion A and the other is labelled as reason R.
Assertion-A :- The gymnosperms are heterosporous.
Reason-R :- Gymnosperms produce two kind of spores, macro (large) and micro (small) spores.
Correct answer from the options given below :

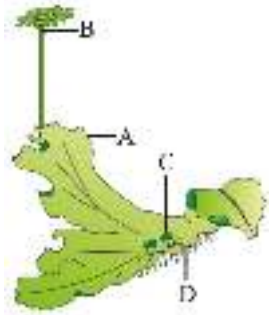
(1) Both A and R are true but R is NOT the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

24) Examine the figure given below and choose the structure in which asexual buds are produced :-



(1) A

(2) B

(3) C

(4) D



25) Identify the following figure of an alga and tell its life cycle pattern :-

(1) *Fucus* - Diplontic
(2) *Fucus* - Haplontic
(3) *Dictyota* - Haplontic
(4) *Porphyra* - Diplontic

26) Identify the given diagram and select the correct option related to that diagram :-



(1) *Pinus* - Possess branched stem
(2) *Cycas* - Possess branched stem
(3) *Pinus* - Pinnate leaves persist for few years
(4) *Ginkgo* - Possess seeds that are cover with ovarian wall

27) Anacardiaceae is a plant family, which of the following belongs to this?

(1) Makoi
(2) Potato
(3) Mango
(4) Wheat

28) Which of the following categories has least similar characters ?

(1) Genus
(2) Species
(3) Family
(4) Class

29) Select the incorrect statement with respect to the taxon, ‘genus’ :-

(1) It is a group or assemblage of related species.

(2) A genus essentially comprises more than one number of species always.

(3) Lion, Tiger, Leopard are closely related species which have been placed in the genus *Panthera*.

(4) *Solanum, Mangifera* and *Homo* are the examples of genera.

30) Which name of potato is correct ?

(1) *Solanum nigrum*

(2) *Solanum melongena*

(3) *Lycopersicon esculentum* Linn.

(4) *Solanum tuberosum* Linn.

31)

Higher the taxonomic category, ____ is the number of organism and ____ is the number of common feature.

(1) Higher, more

(2) Fewer, Less

(3) Higher, Less

(4) Less, More

32) Match the following and choose the correct option:

(i)	Family	(A)	*tuberosum*
(ii)	Kingdom	(B)	Polymoniales
(iii)	Order	(C)	*Solanum*
(iv)	Species	(D)	Plantae
(v)	Genus	(E)	Solanaceae

(1) (i)-D, (ii)-C, (iii)-E, (iv)-B, (v)-A

(2) (i)-E, (ii)-D, (iii)-B, (iv)-A, (v)-C

(3) (i)-D, (ii)-E, (iii)-B, (iv)-A, (v)-C

(4) (i)-E, (ii)-C, (iii)-B, (iv)-A, (v)-D

33) Which is not true for the heterocyst of *Nostoc* :-

(1) Intercalary position

(2) Site of nitrogen fixation

(3) Nitrogenase enzyme present

(4) Photosynthesis

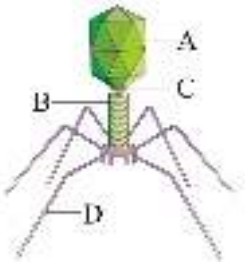
34) Recognise the following figures and find the suitable options :-



(i) A = *Paramoecium*, B = *Euglena*
(ii) Both 'A' and 'B' are Eukaryotes
(iii) A = *Euglena*, B = *Agaricus*
(iv) Both 'A' and 'B' are motile
(v) 'B' is Heterotrophic while 'A' is both autotrophic and some time heterotrophic.

(1) (i), (ii), (iv) & (v)
(2) (i), (iv) & (v)
(3) (ii), (iii) & (v)
(4) (ii), (iii), (iv) & (v)

35) Given below is the diagram of a bacteriophage. In which one of the options all the four parts A, B, C and D are correct ?



	A	B	C	D
1	Tail fibres	Head	Sheath	Collar
2	Sheath	Collar	Head	Tail fibres
3	Head	Sheath	Collar	Tail fibres
4	Collar	Tail fibres	Head	Sheath

(1) 1
(2) 2
(3) 3
(4) 4

36) Identify the organism given in the below figure –



(1) *Euglena* and *Paramoecium*
(2) Slime mould and *Amoeba*
(3) Dinoflagellates and *Chlamydomonas*
(4) Dinoflagellates and Slime mould

37) Read the column-I, II and III and identify the correct match -

	Column-I		Column-II		Column-III
(A)	Sporozoans	(P)	Pseudopodia	(i)	Malaria
(B)	Amaebaid protozoans	(Q)	Flagella	(ii)	Sleeping sickness
(C)	Ciliated protozoans	(R)	Gullet	(iii)	Paramaecium
(D)	Flagellated protozoans	(S)	plasmodium	(iv)	Dysentery

(1) (A)-(S)-(i), (B)-(P)-(ii), (C)-(R)-(iv), (D)-(Q)-(iii)

(2) (A)-(S)-(i), (B)-(P)-(iv), (C)-(R)-(iii), (D)-(Q)-(ii)

(3) (A)-(S)-(ii), (B)-(P)-(iii), (C)-(R)-(i), (D)-(Q)-(iv)

(4) (A)-(S)-(iv), (B)-(P)-(iii), (C)-(R)-(ii), (D)-(Q)-(i)

38) Two statements 1 and 2 are given below :
Statement 1 : Fungi, and members of the Monera and Protista having cell walls have now been excluded from Plantae, though earlier classification given by Whittaker placed them in the same kingdom
Statement 2 : The earliest systems of scientific classification were based mainly on vegetative characters or on the androecium structure. Such systems were artificial and separated the closely related species.

(1) Statement 1 and 2 both are false

(2) Only statement 1 is true

(3) Only Statement 2 is true

(4) Statement 1 and 2 both are true

39) Study the following statements with respect to Bentham and Hooker classification:-
(A) It is largest and natural classification of Spermatophyta.
(B) It was mainly based on floral characters.
(C) It is a simple classification because it is based on actual observation of plants.
(D) In this classification phylogeny is not considered.
How many statement is/are correct?

(1) One

(2) Two

(3) Three

(4) Four

40) How many given below statements are correct statements are?
(A) Diptera is class of Housefly.
(B) Hominidae is an order of Man.
(C) Monocotyledonae is a class of wheat
(D) *Mangifera* is a genus of Mango.
(E) *Musca* is a genus of housefly.

(1) Two

(2) Three

(3) Four

(4) Five

41) Arrange the following taxa for man in ascending sequence :-
(A) Mammalia
(B) Hominidae
(C) Chordata
(D) Primata
(E) *Homo*
(F) Animalia
Options :-

(1) E → B → D → A → C → F

(2) E → C → B → F → D → A

(3) F → C → A → D → B → E

(4) B → D → A → F → C → E

42) Read the following terminology of column-I and II and identify their correct match :

Column-I		**Column-II**	
(A)	Gemmae	(I)	*Fucus*
(B)	Liverwort	(II)	Asexual bud
(C)	Protonema	(III)	*Selaginella*
(D)	Mosses	(IV)	*Marchantia*
		(V)	Sporophyte
		(VI)	*Polytrichum*
		(VII)	Filamentous stage

(1) A-IV, B-III, C-IV, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-VI, B-VII, C-IV, D-II

(4) A-II, B-IV, C-VII, D-VI

43) Read the following statements (A to E) carefully -
(A) The pteridophytes include horstails and mosses.
(B) *Selaginella* is a microphyllous plant.
(C) The gymnosperms are heterosporous.
(D) In *Pinus* and *Cedrus* the stem is branched.
(E) In *Selaginella* sporophylls may form distinct compact structure called strobili or cones.
How may of the above statements are **not** correct?

(1) Two

(2) One

(3) Four

(4) Three

44) From the statements given below choose the correct option :
A. *Polytrichum* have an elaborate mechanism of spore dispersal.
B. *Salvinia* are known as heterosporous pteridophytes.
C. In *Ginkgo*, the male & the female gametophytes have an independent free living existence.
D. In *Salvia* the seeds are enclosed with in a fruits.
E. Both *Salvia* & *Salvinia* are belongs to pteridophytes.

(1) A, B, C are true.
(2) A, B, D are true.
(3) A, B, E are true.
(4) B, D, E are true.

45) Few stages of life cycle of moss is given below. Arrange these events in correct order starting from sex organ bearing structure and select the correct option from option given below
A: Protonema B: Leafy gametophyte
C: Spore formation D: Spore germination
E: Fertilization F: Sporophyte

(1) B → A → C → D → E → F
(2) B → E → D → C → A → F
(3) A → B → E → F → C → D
(4) B → E → F → C → D → A

46) Metagenesis means :-

(1) Polyp produce medusae asexually and medusae produce polyp sexually
(2) Polyp produce medusae sexually and medusae produce polyp sexually
(3) Polyp produce medusae asexually and medusae produce polyp asexually.
(4) Both (1) & (2)

47) Which one is correct about Porifera :-

(1) Unisexual
(2) Internal fertilisation with indirect development
(3) External fertilisation with direct development
(4) External fertilisation with indirect development

48) Which of the following is exclusively marine :-

(1) Sea walnut
(2) *Amphioxus*
(3) *Ascidia*
(4) All above

49) **Statement-I :** Members of cyclostomata are the chordate and are ectoparasite on some fishes.
Statement-II : Gnathostomata has two superclasses, Chondrichthyes & Tetrapoda.

(1) Statement-I & Statement-II are incorrect.
(2) Statement-I correct & Statement-II incorrect.
(3) Statement-I & Statement-II both are correct.
(4) Statement-I incorrect but Statement-II correct.

50) Find the incorrect option regarding to phylum Ctenophora.

(1) They are commonly known as comb-jellies or Sea-walnuts.
(2) They are radially symmetrical.
(3) They are triploblastic organisms with tissue grade body organisation.
(4) Sexes are not separate reproduction take place only by sexual means.

51) Mark the **incorrectly** matched pair :-

(1) *Petromyzon* – Body covered with placoid scales and mouth is ventral
(2) *Labeo* – Two chambered heart
(3) Amphibia – Tympanum represent the ear
(4) Aves – Digestive tract has additional chambers, the crop and gizzard

52)

Match the column-I with column-II and select correct answer about them.

	Column-I		**Column-II**
(a)	Scoliodon	(i)	Hairs, Pinnae and Mammary glands present
(b)	Petromyzon	(ii)	Cold blooded and three chambered heart
(c)	Hemidactylus	(iii)	6-15 pairs of gill slits for respiration
(d)	Pteropus	(iv)	Notochord is persistent throughout life

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(ii)	(i)	(iv)	(iii)

(1) 1
(2) 2
(3) 3
(4) 4

53) **Assertion :** Chondrichthyes have internal fertilisation.
Reason : In Males pelvic fins bear claspers in cartilagenous fish.

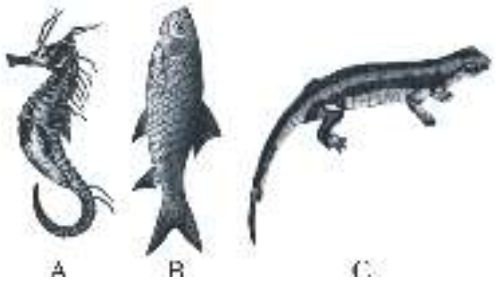
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

54) Refer the figures A, B and C and choose the correct option which shows animals that regulate buoyancy with the help of air bladder:-



(1) A and B

(2) A and C

(3) B and C

(4) B only

55) In cyclostomata, the process of reproduction involves the following events.
A. Adults migrate for spawning to fresh water.
B. The larvae performs metamorphosis in fresh water.
C. After spawning, within a few days, adults die
D. Larvae return to the ocean after metamorphosis
E. Adult cyclostomes lives marine water
Choose the correct sequence of events from the options given below :

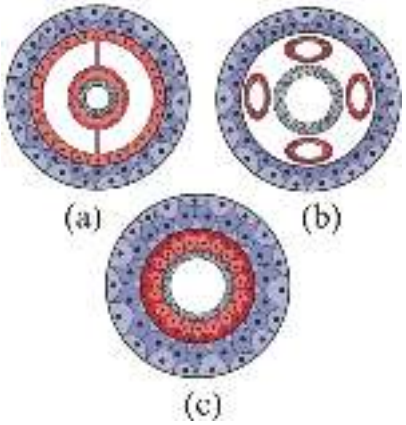
(1) C→E→A→B→D

(2) B→D→C→E→A

(3) E→A→C→B→D

(4) A→E→B→C→D

56) Given below are the diagrams and list of examples find out the correct number of examples with related diagram.



Taenia, Ancylostoma, Pheretima, Anopheles, Chaetopleura, Octopus, Cucumeria, Wuchereria, Fasciola.

(1) a-3, b-5, c-1

(2) a-2, b-2, c-5

(3) a-5, b-2, c-2

(4) a-3, b-3, c-3

57) Read the following statements and select the option which contain all correct statements.
(a) Sponges have a water transport or canal system.
(b) Adamsia is sessile and cylindrical.
(c) Flame cells help in osmoregulation in round worms
(d) Sea - pen and sea-fan both are also known as comb jellies

(1) (a) and (b)

(2) (b) and (c)

(3) (c) and (d)

(4) (a) and (d)

58) Given below is a list of some animals.
Physalia, Pleurobrachia, Gorgonia, Ctenoplana, Aurelia, Sycon, Obelia and *Adamsia*
How many of the above animals are not coelenterates ?

(1) 4

(2) 5

(3) 2

(4) 3

59) Which of the following is a mollusc :-

(1) Sea-horse

(2) Sea-mouse

(3) Sea-hare

(4) Limulus

60) Excretory organ of *Balanoglossus* is :-

(1) Protonephridia

(2) Supra neural gland

(3) Solenocytes

(4) Proboscis gland

61) Which one of the following is a chordate but not a vertebrate :>

(1) *Scoliodon*

(2) *Hag fish*

(3) *Amphioxus*

(4) Star fish

62) A marine cartilaginous fish that can produce electric current is :-

(1) *Pristis*

(2) *Torpedo*

(3) *Trygon*

(4) *Scoliodon*

63) **Assertion :-** Hippocampus is an example of subphylum vertebrata.
Reason :- The members of subphylum vertebrata possess notochord during the embryonic period.

(1) Assertion is correct but reason is incorrect

(2) Both assertion and reason are incorrect

(3) Both assertion and reason are correct but reason is not the correct explanation of assertion

(4) Both assertion and reason are correct but reason is the correct explanation of assertion

64) Match list-I with list-II

	List-I		**List-II**
A	Metagenesis	I	Balanoglossus
B	Radula	II	Pleurobrachia
C	Bioluminescence	III	Obelia
D	Proboscis gland	IV	Dentalium

(1) A–IV, B–II, C–III, D–I

(2) A–II, B–I, C–III, D–IV

(3) A–III, B–IV, C–II, D–I

(4) A–IV, B–I, C–II, D–III

65) Select the incorrect statement about the class Osteichthyes :-

(1) Air bladder is present which helps in buoyancy.

(2) They are mostly viviparous and development is direct.

(3) They have four pairs of gills which are covered by an operculum.

(4) Heart is two - chambered

66) What is primary function of compound Epithelium ?

(1) Absorption

(2) Secretion

(3) Protection

(4) Conduction

67) The inner surface of hollow organ like bronchioles and fallopian tube have :

(1) Squamous epithelium

(2) Cuboidal epithelium

(3) Ciliated epithelium

(4) Compound epithelium

68) Match the columns :-

	Column-I		**Column-II**
A.	Simple epithelium	i.	Some of the columnar or cuboidal cells get specialized for secretion.
B.	Glandular epithelium	ii.	lining for body cavities, ducts and tubes.
C.	Adhering Junction	iii.	Perform cementing to keep heibh ouring cells together
D.	Squamous epithelium	iv.	Made of single thin layer of flattened cells with irregular bondaries.

(1) A–iii, B–ii, C–iv, D–i

(2) A–iv, B–iii, C–i, D–ii

(3) A–ii, B–i, C–iii, D–iv

(4) A–iv, B–i, C–iii, D–ii

69) **Statement I :-** Connective tissue are most abundant and widely distributed in the body of complex animals.
Statement II :- Connective tissues range from soft connective tissue to specialized types which include cartilage, bone adipose and blood.
Choose the correct option from the given below.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct.

70) **Statement I :-** The squamous epithelium is made up of a single thin layer of flattened cells with irregular boundaries.
Statement II :- Endothelium is the inner lining of blood vessels and lymph vessels.
Choose the correct option from the given below.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct.

71) ..... present on free surface of columnar epithelium of stomach and intestine.

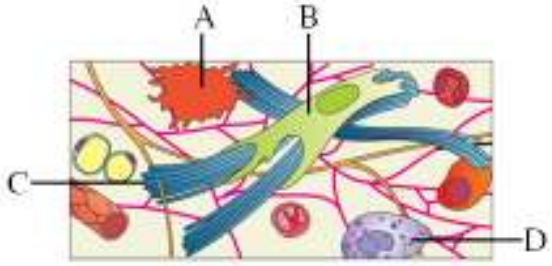
(1) Basal nuclei

(2) Microvillli

(3) Cilia

(4) Irregular bondaries

72) The labelling A, B, C, D of the following diagram respectively :-



	A	B	C	D
(1)	Fibroblast	Mast cell	Collagen fibre	Macrophage
(2)	Macrophage	Fibroblast	Collagen fibres	Mast cell
(3)	Fat storage area	Mast cell	Macrophage	Fibroblast
(4)	Macrophage	Collagen fibre	Elastin fibre	Mast cell

(1) 1

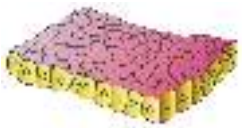
(2) 2

(3) 3

(4) 4

73)

Consider the following figure and choose the correct option :-



	Tissue	Site	Function
(1)	Squamous epithelium	Found in blood vessels	Forms diffusion surface
(2)	Cuboidal epithelium	Found in ducts of glands	Main function is secretion and absorption
(3)	Ciliated columnar epithelium	Trachea	Movement of mucus
(4)	Columnar epithelium	Lining of stomach	Absorption

(1) 1

(2) 2

(3) 3

(4) 4

74) In cockroach what is number and location of spiracles respectively ?

(1) 10 and dorsal side of body

(2) 10 and lateral side of body

(3) 10 pairs and ventral side of body

(4) 10 pairs and lateral side of body

75) Which of the following excretory structure absent in female cockroach ?

(1) Malpighian tubule
(2) Urecose gland
(3) Fat body
(4) Nephrocyte cell

76) In cockroach, the sense organs are :

(1) Antennae
(2) Maxillary palps
(3) Labial palps
(4) All of these

77) Find out the incorrect statement :

(1) Cockroaches are serious pests and vectors of several diseases
(2) Cockroaches are nocturnal, herbivores animals
(3) Cockroaches are included in class insecta of phylum arthropoda
(4) Common species of cockroach is *Periplaneta americana*.

78) Match **List–I** with **List-II** regarding the organs of Cockroach :

List–I		**List–II**	
(a)	Crop	(i)	grinding the food particles
(b)	Proventriculus	(ii)	secretion of digestive juice
(c)	Hepatic caecae	(iii)	removal of nitrogenous waste
(d)	Malpighian tubules	(iv)	storage of food

Choose the **correct answer** from the options given below :
(1) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
(2) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)
(3) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
(4) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)

79) **Statement - I :** Cockroach eyes consist of several ommatidia. So they receive several images of an object.
Statement - II : Mosaic vision is present in cockroach with more sensitivity but less resolution.

(1) Both statement I and statement - II are incorrect.
(2) Statement - I is correct but statement - II is incorrect.
(3) Statement - I is incorrect but statement - II is correct.
(4) Both statement - I and statement - II are correct.

80) Read column-I, II & III and identify their correct match :-

Column-I		Column-II		Column-III	
(A)	Anal cerci	(P)	On 9th sternum	(i)	Sound receptor
(B)	Anal style	(Q)	On metathorax	(ii)	Sperm storage
(C)	Hind wings	(R)	On 10th tergum	(iii)	Helps in flying
(D)	Spermatheca	(S)	6th abdominal segment	(iv)	Copulation

(1) (A)-(R)-(i), (B)-(P)-(iv), (C)-(S)-(iii), (D)-(Q)-(ii)

(2) (A)-(R)-(i), (B)-(S)-(iv), (C)-(P)-(iii), (D)-(Q)-(ii)

(3) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(4) (A)-(R)-(i), (B)-(P)-(iv), (C)-(Q)-(iii), (D)-(S)-(ii)

81) Read following terminology of column-I & II and identify their correct match :-

Column-I		Column-II	
(A)	Tergum	(i)	Help in monitoring the external environment
(B)	Sternum	(ii)	Sclerites of dorsal side
(C)	Antennae	(iii)	Grinding of food
(D)	Gizzard	(iv)	Sclerites of ventral side
		(v)	Jaws of cockroach
		(vi)	Copul along structure
		(vii)	Helps in flying

(1) (A)-(iv), (B)-(ii), (C)-(vi), (D)-(iii)

(2) (A)-(ii), (B)-(iv), (C)-(iii), (D)-(vii)

(3) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)

(4) (A)-(iv), (B)-(ii), (C)-(iii), (D)-(v)

82) In frogs, heartbeat is controlled by :-

(1) Cerebellum

(2) Spinal cord

(3) Medulla oblongata

(4) Mid-brain

83) Which one of the following is common in both male and female frog ?

(1) Bidder's canal

(2) Vocal sac

(3) Cloaca

(4) Copulatory pad

84) **Assertion:** Duodenum receives bile from gall bladder and pancreatic juice from the pancreas

through a common duct in frog.
Reason: Salivary glands are absent in frog, So digestion start in stomach.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

85) In frog, cutaneous respiration takes place

(1) Always
(2) Only on land
(3) Only in water with pulmonary respiration
(4) Only in water, pulmonary respiration is not occurring

86) Read the following statements carefully and choose the correct option.
Statement-1 : In frog, a triangular structure called sinus venosus joins the right atrium.
Statement-2 : In frogs, the elimination of nitrogenous wastes is carried out by a well developed excretory system, which consist of a pair of kidneys, ureters, one cloaca and one urinary bladder.

(1) Both the statements (1) and (2) are correct.
(2) Statement (1) is correct but statement (2) is incorrect.
(3) Statement (2) is correct but statement (1) is incorrect.
(4) Both the statement (1) and (2) are incorrect.

87) Following are given four statements (A) to (D) related to frog, read the statements carefully and select the option that identifies two correct statements.
A. The ovaries are situated near kidney and there is no functional connection with kidneys.
B. Vasa efferentia enter the kidneys on their side and open into bidder's canal.
C. There are twelve pairs of cranial nerves arising from the brain
D. RBCs are enucleated and contain red coloured pigment.

(1) A and C
(2) A and B
(3) C and D
(4) A and D

88) Match List-I with List-II :

	List-I		**List-II**
(A)	Vasa efferentia	(I)	2500-3000
(B)	Ova	(II)	Summer sleep
(C)	Hibernation	(III)	10-12
(D)	Aestivation	(IV)	Winter sleep

Choose the correct answer w.r.t. frog from the options given below :

(1) (A–I), (B–III), (C–IV), (D–II)

(2) (A–III), (B–I), (C–II), (D–IV)

(3) (A–III), (B–I), (C–IV), (D–II)

(4) (A–I), (B–III), (C–II), (D–IV)

89) **Assertion:** Heart of frog is muscular and three chambered.
Reason: Heart of frog pumps mixed blood into lungs and body by or with the help of conus arteriousus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

90) Which of the following statements are correct in reference with the frog ?
A. Eyes are bulged and covered by nictitating membrane.
B. Membranous tympanum receives the sound signals.
C. The frog never drinks water
D. A pair of nostrils is present above the mouth.

(1) A and B

(2) C and D

(3) A and D

(4) A,B,C and D

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	3	2	1	2	3	4	3	2	4	3	3	4	2	3	2	4	3	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	2	4	2	1	4	4	1	2	2	4	2	1	1	4	3	4	1	1
Q.	41	42	43	44	45															
A.	4	3	3	1	3															

CHEMISTRY

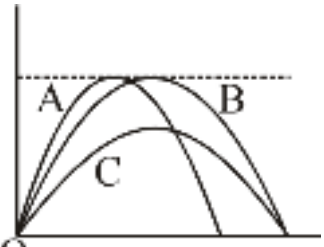
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	2	1	1	2	4	3	2	4	2	2	1	1	2	2	4	3	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	3	3	1	2	2	2	3	2	1	2	1	4	1	4	2	3	4	1
Q.	86	87	88	89	90															
A.	2	2	2	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	1	2	3	1	4	2	3	4	1	4	4	1	1	1	3	4	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	4	3	1	1	3	4	2	4	3	2	4	3	3	4	2	3	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	2	2	4	1	2	4	2	3	1	2	1	1	3	3	1	4	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	3	3	2	3	3	3	4	4	2	2	2	4	2	4	2	1	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	3	2	1	1	2	3	2	4										

SOLUTIONS

PHYSICS



1)
(A) time of flight (III) equal for A & B
(B) u_y/u_x (I) greater for A only
(C) u_x (II) greater for C only
(D) u_xu_y (IV) equal for B & C

2) $v^2 x^3 = 4$

$2v\frac{dv}{dt}x^3 + 3x^2\frac{dx}{dt}v^2 = 0$; $v^2x^3 = 4$

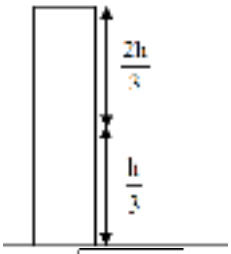
$vx^2\left(2x\frac{dv}{dt} + 3v^2\right) = 0$; $v = \left(\frac{4}{x^3}\right)^{\frac{1}{2}}$

$\frac{dv}{dt} = \frac{-3v^2}{2x}$

$\frac{dv}{dt} = \frac{-3v^2v^{2/3}}{2(4)^{1/3}} \rightarrow a = \frac{-3v^{8/3}}{2^{5/3}}$

3) $V_{rel}^2 \geq 2a_{rel}x$

4) Let time to reach may height = T = $\sqrt{\frac{2h}{g}}$



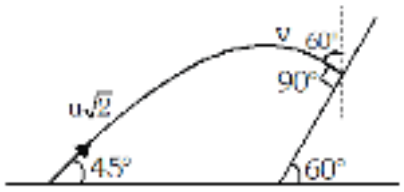
$t = \sqrt{\frac{2}{3} \times \frac{2h}{g}} = \frac{\sqrt{2}}{\sqrt{3}}T$

$t_1 = T - t$

$t_2 = T + t$

$\frac{t_1}{t_2} = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

5)



$\left(u\sqrt{2}\right)\cos 45^\circ = v\sin 60^\circ$

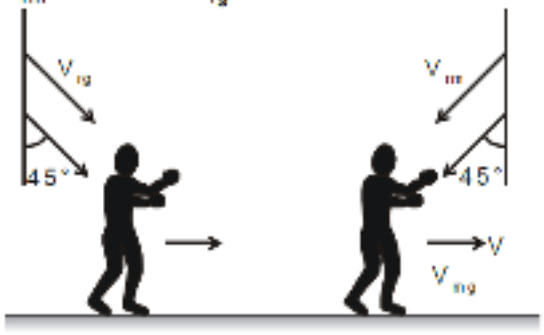
$u = \frac{v\sqrt{3}}{2}$, $v = \frac{2u}{\sqrt{3}}$

6) $100 = \frac{1}{2} \times 0.5 \times t^2$

7) $\vec{V}_{rg} = \vec{V}_{rm} + \vec{V}_{mg}$

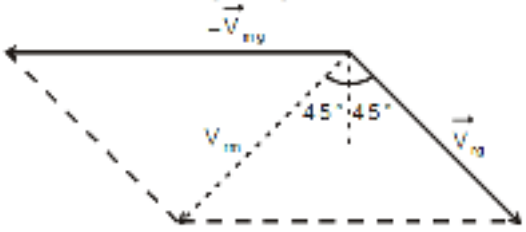
$\vec{V}_{rm} = \vec{V}_{rg} - \vec{V}_{mg}$

$V_{rm}\cos 45^\circ = V_{rg}\cos 45^\circ$



$V_{rm} = 2\sqrt{2}$ m/s $= V_{rg}$

$V_{rm}\cos 45^\circ = V_{mg} - V_{rg}\cos 45^\circ$



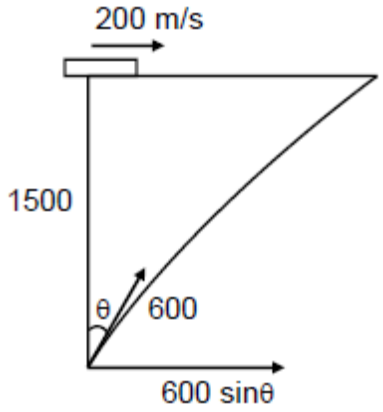
$V_{mg} = 2\sqrt{2}\frac{1}{\sqrt{2}} + 2\sqrt{2}\frac{1}{\sqrt{2}} = 4$ m/s

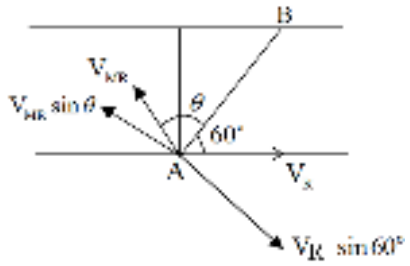
s = 16 m.

8) $600\sin\theta = 200$

$\theta = \sin^{-1}\left(\frac{1}{3}\right)$

n = 3





9)

for swimmer to move along AB not velocity

$\perp$ to AB must be zero.

$V_{MR} \sin\theta - V_R \sin 60^\circ = 0$

$15 \sin\theta = 10 \sqrt{3} \times \frac{\sqrt{3}}{2}$

$\sin 60° = \frac{100\sqrt{3}}{AB}$ = AB = 200 m

$\sin\theta = 1$

$\theta = 90^\circ$

$t = \frac{AB}{V_{MR}\cos\theta + V_R \cos 60^\circ}$

$t = \frac{200}{0 + 10\sqrt{3}\cos 60^\circ} = \frac{200}{5\sqrt{3}}$

$= \frac{40}{\sqrt{3}}$

$\Rightarrow n = 3$

10)

Range & Speed are equal

$\theta, \theta + 5$

$2\theta + 5 = 90$

$2\theta = 85$

$\theta = \frac{85}{2} = 42.5^\circ$

11) $\left|\vec{V}_{avg}\right| = \frac{|\text{displacement}|}{\text{time}}$

$= \frac{\frac{\pi(1)^2}{2}}{2}$

= π/4 m/s

12) $x = u_x t + \frac{1}{2} a_x t^2$

$y = u_y t + \frac{1}{2} a_y t^2$

$32 = 0 \times t + \frac{1}{2}(4)(t)^2$

$t^2 = 16$

t = 4 sec

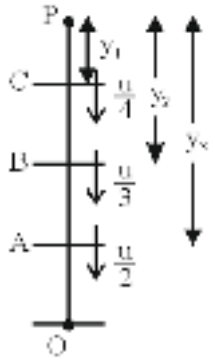
$x = 3 \times 4 + \frac{1}{2} \times 6 \times 4^2$

= 12 + 48 = 60 m

13)

$a_x = 2$, $a_y = 0$, $v_x = 2t + 2$, $v_y = 2$

$\vec{a}$ and $\vec{v}$ are non-parallel and $\vec{a}$ = constant



14)

$\frac{u}{2} = \sqrt{2gy_3} \Rightarrow y_3 = \frac{\mu^2}{8g}$

in similar way

$y_2 = \frac{\mu^2}{18g}$

$y_1 = \frac{\mu^2}{32g}$

$\frac{AB}{AC} = \frac{y_3 - y_2}{y_3 - y_1} = \frac{20}{27}$

15) (P) 1 MSD = 1mm

10 VSD = 9 MSD

LC = 1MSD –1 VSD

= 1mm - 0.9 mm = 0.1 mm (C)

(Q) 1 MSD = 0.5 mm

10 VSD = 9 MSD

LC = 1MSD – 1VSD

$= 0.5 \text{ mm} - \frac{9(0.5\text{mm})}{10}$ =0.5 x 0.1 = 0.05 mm (A)

(R) 1 MSD = 0.5 mm

20 VSD = 19 MSD
LC = 1MSD – 1 VSD
$= 0.5\text{ mm}\left(1-\frac{19}{20}\right) \Rightarrow \frac{0.5}{20} = 0.025\text{mm}$ (D)
(S) 1MSD = 1mm
100 VSD = 99 MSD
LC = 1MSD – 1VSD
$= 1\text{ mm} - \frac{99}{100}\text{mm} = 1 - 0.99 = 0.01\text{ mm}$ (B)

16)

Least count of screw gauge
$= \frac{\text{Pitch}}{\text{Number of circular divisions}}$
$= \frac{0.5\text{ mm}}{50} = 0.01\text{ mm}$
$\Delta D = 0.01\text{ mm}$

Diameter $D = 2.5\text{ mm} + 20 \times \frac{0.5}{50}$
$D = 2.70\text{ mm}$
density $d = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{m}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3}$
Percentage error in d
$\left(\frac{\Delta d}{d}\times 100\right)\% = \left(\frac{\Delta m}{m}\times 100\right)\% + 3\left(\frac{\Delta D}{D}\times 100\right)\%$
$= 2\% + 3\left(\frac{0.01}{2.7}\times 100\right)\%$
$= 2\% + \frac{3}{2.7}\%$
$= 2\% + 1.11\% = 3.11\%$

17) $\vec{A} = 2\hat{i} + 3\hat{j} \qquad \vec{B} = -\hat{i} + 2\hat{j}$
$\vec{C} = -3\hat{i} - \hat{j} \qquad \vec{D} = +2\hat{i} - 2\hat{j}$
$\vec{A} + \vec{B} + \vec{C} + \vec{D} = 2\hat{j} \uparrow$

18) $\alpha\bar{A} = \alpha\hat{i} + \alpha\hat{j} - 2\alpha\hat{k}$
$\beta\bar{B} = \beta\hat{i} - \beta\hat{j} - \beta\hat{k}$
$(\alpha\bar{A} + \beta\bar{B}).\bar{C} = 0$
$= \left(\hat{i}(\alpha+\beta) + \hat{j}(\alpha-\beta) - \hat{k}(2\alpha+\beta)\right).\left(2\hat{i} - 3\hat{j} - 4\hat{k}\right) = 0$
$\Rightarrow 2\alpha + 2\beta - 3\alpha + 3\beta + 8\alpha + 4\beta = 0$
$7\alpha + 9\beta = 0$
$\frac{\alpha}{\beta} = -\frac{9}{7}$

19) $\frac{dy}{dx} = 3\cos 3x - 4\sin 3x$

so $\left.\frac{dy}{dx}\right|_{max} = \sqrt{3^2 + 4^2} = 5$

20) Standard results of graph

21) $y = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x} + 1\right)$

$y = x^2 - 1 + x + 1 - \frac{1}{x^2} + \frac{1}{x}$

$\frac{dy}{dx} = 2x + 1 + \frac{2}{x^3} - \frac{1}{x^2}$

22) $D = 2 \times 1 + 5 \times \frac{10-9}{100} = 2.05$ cm

23) $y = \frac{MgL^3}{4bd^3\delta}$

$\frac{\Delta y}{y} = \frac{\Delta M}{M} + \frac{3\Delta L}{L} + \frac{\Delta b}{b} + \frac{3\Delta d}{d} + \frac{\Delta\delta}{\delta}$

$\frac{\Delta y}{y} = \frac{10^{-3}}{2} + \frac{3\times 10^{-3}}{1} + \frac{10^{-2}}{4} + \frac{3\times 10^{-2}}{4} + \frac{10^{-2}}{5}$

$= 10^{-3}[0.5 + 3 + 2.5 + 7.5 + 2] = 0.0155$

24) In air $w_1 = 15.0 \pm 0.3$ N

In water $w_2 = 10.0 \pm 0.2$ N

$W_1 = \rho vg$

Buoyant force $B - \rho_w vg = w_1 - w_2$

Relative density $= \frac{\rho}{\rho w} = \frac{w_1}{B}$

$\frac{\Delta(RD)}{RD} \times 100 = \frac{\Delta w_1}{w_1} \times 100 + \frac{\Delta B}{B} \times 100$

$B = (15.0 \pm 0.3) - (10.0 \pm 0.2)$

$B = 5.0 \pm 0.5$ N

$\frac{\Delta(RD)}{RD} \times 100 = \pm\left(\frac{0.3}{15.0} \times 100 + \frac{0.5}{5.0} \times 100\right)$

$= \pm(2 + 10) = \pm 12\%$

25)

$\Delta\ell = 1$ mm

N = 50 division

zero error = + 6 Divisions

= + 0.12 mm

Diameter = measured value + zero correction

$= 3mm + (31 - 6) \times \frac{1}{50}\ mm$

$= 3mm + 0.50mm = 3.50\ mm$

26) $w_{friction} = \Delta KE + \Delta U$

$-\mu mg(\ell + x) = -\frac{1}{2}mu^2 + \frac{1}{2}kx^2$

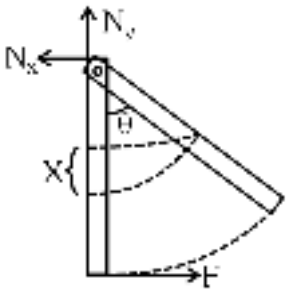
$-0.1 \times 50 \times (10.5) = \frac{1}{2}(5)(20)^2 + \frac{1}{2}K(0.5)^2$

$\frac{1}{8}K = 1000 - 52.5 = 947.5$

$\boxed{K = 7,580\ N/m}$

27) $\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k} = -\left[4x\hat{i} + 9y^2\hat{j} + 2\hat{k}\right]$

28)



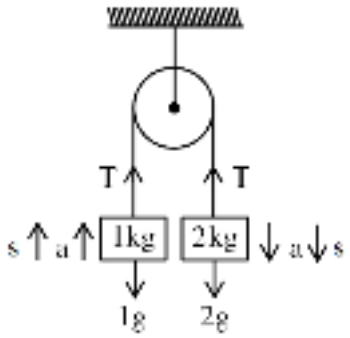
$W_g + W_{N_X} + W_Y + W_F = \Delta KE$

$-mg\left(\frac{\ell}{2} - \frac{\ell}{2}\cos 60\right) + 0 + 0 + W_f = 0$

$W_f = mg\frac{\ell}{2}\ \frac{\ell}{2}$

$= 12.5\ J$

29)



$a = \left(\frac{2-1}{2+1}\right)(10) = \frac{10}{3}m/s^2$

$T = \frac{2m_1m_2}{m_1 + m_2}g$

$T = \frac{2(1)(2)}{3}(10)$

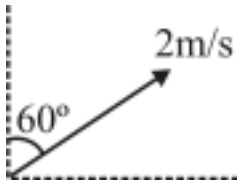
$= \frac{40}{3}N$

and displacement. in t = 0.3 sec.

$$s = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{10}{3}\right)\left(\frac{3}{10}\right)^2 = 0.15$$

Work done by gravity on 2 kg block = +mgh = mgs = +3 J
Work done by gravity on 1 kg block = –mgh = –mgs = –1.5 J
Work done by tension (string) on 2 kg block = –Ts = –2 J [θ = 180º]
Work done by tension (string) on 1 kg = Ts [θ = 0º] $= \left(\frac{40}{3}\right)(0.15) = 2J$
a→r, b→p, c→s, d→q

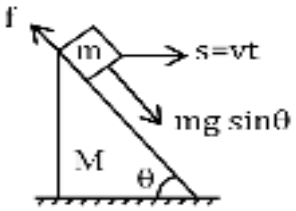
30)



P = Fvcosθ = mg vcosθ

$= 50 \times 9.8 \times 2 \times \frac{1}{2} = 490$ W

31)



As the block is rest with respect to the wedge, friction force f = mg sin θ

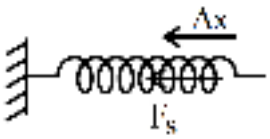
work done by friction
W = fs cos(180° – θ)

W = mg sin θ (vt) (–cos θ) = – mg (sinθ cosθ) vt

$$W = -\frac{mg\sin 2\theta}{2}vt$$

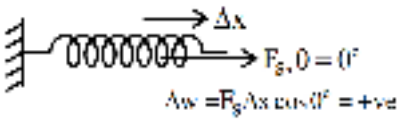
32)

When a spring returns to its natural length
Case-I from elongated position



$\Delta W = F_s \Delta x \cos 0^\circ$ – ve

Case-II from compressed position

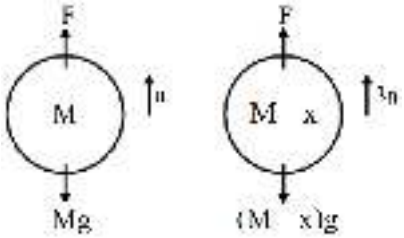


33) $P_{in} = \frac{mgh}{t} = \frac{10 \times 10 \times 100}{1} = 10\,kW$

P_{out} = 80% of P_{in}

= 8 kW

34)



F – Mg = Ma

F = Ma + Mg

F – (M – x)g = (M – x)3a

Put F

Ma + Mg – Mg + xg = 3Ma – 3xa

$x = \frac{2Ma}{g + 3a}$

35)

Limiting value of friction on pen is $f_1 = \mu_1 mg$

So, for pen to start slipping, force acting over pen is

$F_1 \geq f_1 = \mu_1 mg$

So, minimum acceleration of pen is $a_{min} = \frac{F_1}{m} = \mu_1 g$

Hence, minimum acceleration of paper is $a_{min} = \mu_1 g$

Now, consider free body diagram of paper

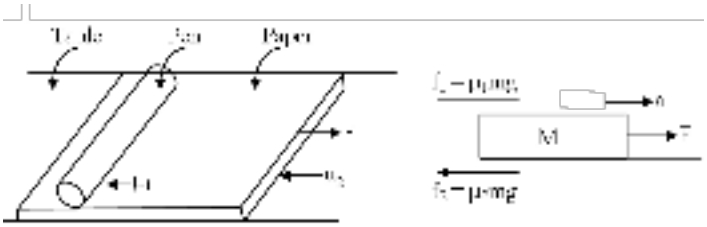
We have,

$F_{net} = Ma_{min} = F - f_1 - f_2$

$\Rightarrow F = Ma_{min} + f_1 + f_2$

$= M\mu_1 g + \mu_1 mg + \mu_2 (m + M)g$

$= (M + m)(\mu_1 + \mu_2)\, g$



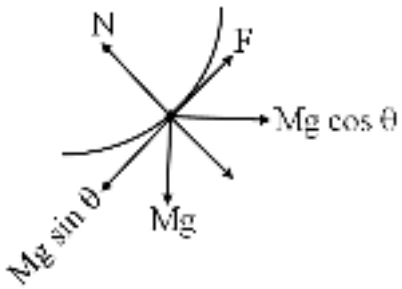
36) $\tan\theta = \mu, \frac{dy}{dx} = \frac{x^2}{2}$ (from question)

coefficient of friction μ = 0.5

$\square\ 0.5 = \frac{x^2}{2}$

$\Rightarrow x = \pm 1$

Now, $y = \frac{x^3}{6} = \frac{1}{6}m$



$F = \mu N$

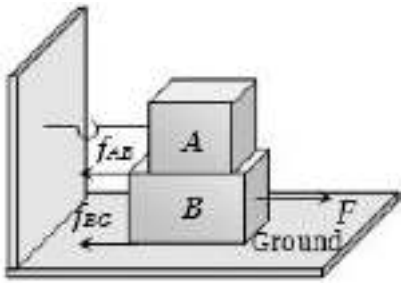
$Mg \sin\theta = \mu\, mg \cos\theta$

37)

$F = f_{AB} + f_{BG}$

$= \mu_{AB} m_A g + \mu_{BG} \times (m_A + m_B) g$

$= 0.2 \times 100 \times 10 + 0.3(300) \times 10 = 200 + 900 = 1100$ N



38)

Block will not slip if

$(m_1 + m_2)\, g\sin\theta \le \mu m_2 g\cos\theta$

$3 \sin\theta \le \left(\frac{3}{10}\right)(2)\cos\theta$

$\tan\theta \leqslant \frac{1}{5} \Rightarrow \theta \leqslant 11.5^{\circ}$

(P)	θ = 5°	friction is static	$f = (m_1 + m_2)\, g\sin\theta$
(Q)	θ = 10°	friction is static	$f = (m_1 + m_2)\, g\sin\theta$
(R)	θ = 15°	friction is kinetic	$f = \mu m_2 g\cos\theta$
(S)	θ = 20°	friction is kinetic	$f = \mu m_2 g\cos\theta$

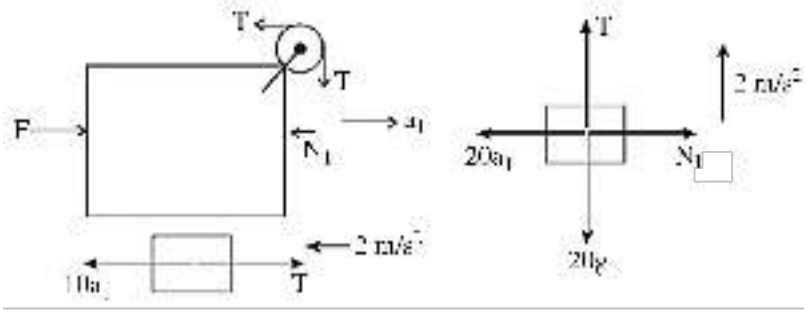
39)

Let acceleration of 100 kg block = a_1 FBD of 100 kg block w.r.t ground

$F - T - N_1 = 100a_1$ .....(i)

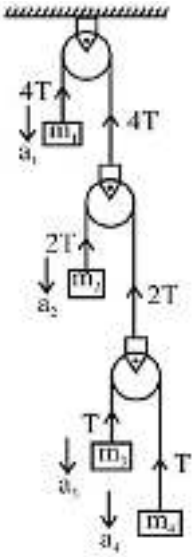
FBD of 20 block wrt 100 kg

T – 20g = 20(2)
T = 240
$N_1 = 20a_1$
FBD of 10 kg block wrt 100 kg
$10a_1 – 240 = 10(2)$
$a_1 = 26\ m/s^2$
F – 240 – 20(26) = 100 × 26
⇒ F = 3360 N



40)

By method of virtual work
$\sum \vec{T} . \vec{a} = 0$
$–4Ta_1 – 2Ta_2 – Ta_3 – Ta_4 = 0$
$4a_1 + 2a_2 + a_3 + a_4 = 0$



41) The sand bag is placed on a trolley that is moving with a uniform speed of 27 km/h. The external forces acting on the system of the sandbag and the trolley is zero. When the sand starts leaking from the bag, there will be no change in the velocity of the trolley. This is because the leaking action does not produce any external force on the system. This is in accordance with Newton's first law of motion. Hence, the speed of the trolley will remain 27 km/h.

42)

Force = Rate of change of momentum

Initial momentum

$\vec{P}_1 = mv\sin\theta\hat{i} + mv\cos\theta\hat{j}$

Final momentum

$\vec{P}_2 = -mv\sin\theta\hat{i} + mv\cos\theta\hat{j}$

$\vec{F} = \frac{\Delta\vec{P}}{\Delta t} = \frac{-2mv\sin\theta}{2\times10^{-3}}$

Substituting m = 0.1 kg, v = 5 m/s, θ = 60°

Force on the ball $\vec{F} = -250\sqrt{3}$ N

Negative sign indicates direction of the force

43)

Impulse = Change in momentum = $m(v_2 - v_1)$ ....(i)

Again impulse = Area between the graph and time axis

$= \frac{1}{2}\times 2\times 4 + 2\times 4 + \frac{1}{2}(4+2.5)\times 0.5 + 2\times 2.5$

$= 4 + 8 + 1.625 + 5 = 18.625$ ....(ii)

From (i) and (ii), $m(v_2 - v_1) = 18.625$

$\Rightarrow v_2 = \frac{18.625}{m} + v_1 = \frac{18.625}{2} + 5 = 14.31$ m/s

44) When the spring C is stretched slowly, the tension in A is greater than that of C, because of the weight mg and the former reaches breaking point earlier.

45)

When the bird flies, it pushes air down to balance its weight.

So the weight of the bird and closed cage assembly remains unchanged.

CHEMISTRY

46) For WASB type of salts

$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$

For weakest acid pK_a will be highest and K_a will be lowest.

47)

$h = \sqrt{\frac{K_W}{k_a\times C}}$

$\Rightarrow 10^{-4} = \sqrt{\frac{10^{-14}}{10^{-6}\times C}}$

$\Rightarrow C = 1 \quad \square [NaCN] = 1 = \frac{W}{MM}\times\frac{1}{V_L}$

$\Rightarrow 1 = \frac{w}{49} \times \frac{1}{0.5} \Rightarrow w = 24.5$ g

48) NaCN + HCl → NaCl + HCN

4m	2m	0	0
4–2		2m	2m
2m			

$[H^+] = K_a \times \frac{[HCN]}{[CN^-]}$

$[H^+] = 4 \times 10^{-6} \times \frac{2}{2}$

$[H^+] = 4 \times 10^{-6}$

pH = 6 – log4 = 6 – 0.6 = 5.4

Final answer : 2

49)

	NH_3	+ HCl	→ NH^+_4	+ Cl^-
m-mol	10	5	0	0
	5	0	5	5

Basic buffer

50) **Concept :** This question is based on for a salt $M_aX_b \rightarrow aM^+ + bX^-$, $K_{sp} = a^a \times b^b \times s^{(a+b)}$

Solution :

(a) $Bi_2S_3 \rightarrow 2Bi^{3+} + 3S^{2-} \rightarrow K_{sp} = 108\ s^5 \rightarrow$ (iii) (b) $CdS \rightarrow Cd^{2+} + S^{2-} \rightarrow K_{sp} = s^2 \rightarrow$ (iv)

(c) $Al(OH)_3 \rightarrow Al^{3+} + 3OH^- \rightarrow K_{sp} = 27\ s^4 \rightarrow$ (ii)

(d) $CaF_2 \rightarrow Ca^{2+} + 2F^- \rightarrow K_{sp} = 4s^3 \rightarrow$ (i)

Code (a) (iii), (b) (iv), (c) (ii), (d) (i)

Final answer : 1

51) I.P. = $(0.01)\ (10^{-3})^2 = 10^{-8} > K_{sp}$

52)

Millimoles of $H^+ = 10^{-2} \times 200 = 2$

Millimoles of $OH^- = 10^{-2} \times 300 = 3$

$[OH^-]_{mix} = \frac{3-2}{200+300} = 2 \times 10^{-3}$

pH = 11.3

53) $pH = \frac{1}{2}(pKa - \log C) = \frac{1}{2}(4 - (-1)) = 2.5$

$pH = pKa + \log \frac{[A^-]}{[HA]} = 4 + 0 = 4$

$pH = 7 + \frac{1}{2}(pKa + \log C)$

$= 7 + \frac{1}{2}(4 + \log\frac{2}{30}) = 8.41$

54) $[OH^-] = \sqrt{K_b.C}$

55)

Moles of HCl = 0.05 L × 0.2 M = 0.01 mol
Moles of CH_3COOH = 0.05 L × 0.2 M = 0.01 mol
Total volume = 0.1 L
$[H^+]$ from HCl = 0.01 / 0.1 = 0.1 M
pH = −log(0.1) = 1.0
(CH_3COOH ionisation is suppressed due to common H^+ ion.)
Final Answer : (4)

56) **Solution :**
Assertion (A) : True, small amounts of strong acid/base change pH of water.
Reason (R) : Incorrect. Adding acid/base does not increase water's ionisation; it suppresses it (common ion effect).
Final Answer : 2

57) $N_2O_4 \rightleftharpoons 2NO_2$
1 0
At eq^n (1 – α) (2α total = 1 + α)

$K_P = \frac{\left(\frac{2\alpha}{1+\alpha} \times P\right)^2}{\left(\frac{1-\alpha}{1+\alpha} \times P\right)}$

$K_P = \frac{4\alpha^2.P}{1-\alpha^2}$

$K_P - K_P\,\alpha^2 = 4\alpha^2.P$

$\alpha^2 (4P + K_P) = K_P$

$\alpha = \sqrt{\frac{K_p}{4p + K_p}}$

58)

$K_c = \frac{(0.2)^2\,(0.1)}{(1)\;\;(2)^2} = 0.001$

59) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
initially 1 3 0
at eq^n (1–0.5) (3–1.5) 2 × 0.5 = 1 mole
 Total moles at equilibrium = 0.5 + 1.5 + 1 = 3

Partial pressure of NH_3 = $P \times \left(\frac{1}{3}\right) = \frac{P}{3}$

60)

Solution :
For the reaction ; $H_2 + I_2 \rightleftharpoons 2HI$
Given: $[H_2] = 2$ M, $[I_2] = 2$ M, $[HI] = 8$ M, $K_c = 4$
$Q_c = (HI)^2 / (H_2 \times I_2) = (8)^2 / (2 \times 2) = 64 / 4 = 16$
Given $K_c = 4$
Since $Q_c > K_c \rightarrow$ backward direction
So, Reaction proceeds in backward direction
Final Answer : option (2)

61) **Solution:**
Because liquid water is denser than ice, it occupies a smaller volume. According to Le Chatelier's principle, increasing the pressure on the system will shift the equilibrium to favor the denser, lower-volume phase, which is liquid water. This shift means that more ice will melt to form water.
Final Answer: (2)

62) NCERT 2023-24 Pg # 186
Oxalic acid will produce $C_2O_4^{2-}$ ions which will combine with Fe^{+3} to form complex $[Fe(C_2O_4)_3]^{3-}$.
So concentration of $Fe^{+3}\downarrow$ and equilibrium will shift backward.
KSCN will reproduce SCN^-, equilibrium will shift forward, so intensity of red colour will increase $HgCl_2$ will produce Hg^{+2} which will combine with SCN^- to form $[Hg(SCN)_4]^{2-}$. So equilibrium will shift backward.

63) $2Fe(NO_3)_3 + 3Na_2CO_3 \rightarrow Fe_2(CO_3)_3 + 6NaNO_3$
mol 3 4
Limiting reagent is Na_2CO_3 and moles of $NaNO_3$ formed = $4 \times 2 = 8$
% yield = $\frac{6.4}{8} \times 100 = 80$

64)

$MgCO_3 \rightarrow MgO + CO_2$
1 mol 1 mol
84 g 40 g
by 84 g ___ MgO = 40 g
20g ___ MgO = $\frac{40}{84} \times 20 = \frac{800}{84}$g
% Purity = $\frac{8}{\left(\frac{800}{84}\right)} \times 100 = 84\%$

65) **Explain Question:** Match amounts using the reaction: $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$.
Concept: This question is based on mass-volume relationship
Solution:
(a) 8 mol $H_2O \Rightarrow C_3H_8$ needed = 8/4 = 2 mol ⇒ volume = 2×22.4 = 44.8 L ⇒ (iii).
(b) 88 g C_3H_8 = 2 mol ⇒ O_2 needed = 2×5 = 10 mol ⇒ mass = 10×32 = 320 g ⇒ (iv).
(c) 44 g C_3H_8 = 1 mol ⇒ CO_2 formed = 3 mol ⇒ (ii).
(d) 1 mol C_3H_8 with 5 mol $O_2 \Rightarrow H_2O$ = 4 mol ⇒ mass = 4×18 = 72 g ⇒ (i).
(a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
Final Answer: 4

66)

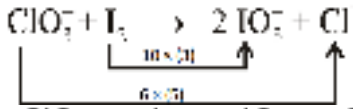
Element	%	At. weight	% Atomic weight	Simplest molar ratio
C	54.55	12	$\frac{54.55}{12} = 4.54$	$\frac{4.54}{2.27} = 2$
H	9.09	1	$\frac{9.09}{1} = 9.09$	$\frac{9.09}{2.27} = 4$
O	36.36	16	$\frac{36.36}{16} = 2.27$	$\frac{2.27}{2.27} = 1$

 Empirical formula is C_2H_4O.

67) **Concept:** Atomic mass unit (u)
Solution:
1 u = 1/5 × Mass of C-12 atom
1 u = 1/5 ×12 nucleons
Mass of 1 nucleons = 5/12 u In Nitrogen atom 14 nucleons are present atomic mass of N = (5/12)×14 = 5.83 u = 5.83amu
Final Answer: Option 2

68) +7 +4 +2 +6
$MnO_4^- + SO_3^{2-} \rightarrow Mn^{+2} + SO_4^{2-}$
 equivalents of O.A. = equivalents of R. A.
or 1 × 5 = n × 2
 $n = \frac{5}{2}$ = 2.5 mol.

69) **Explanation Question:**
To find out sum of S. C. of H_2O and H^+.
Concept:
Balancing of Redox reactions.
Solution:
$\overset{-5}{ClO_3^-} + \overset{0}{I_2} + XH_2O \rightarrow \overset{+5}{IO_3^-} + \overset{-1}{Cl^-} + YH^+$



$5ClO_3^- + 3I_2 \rightarrow 6IO_3^- + 5Cl^- + 6H^+ + 3H_2O$

X = 3, Y = 6,

Final answer : Option (3)

70) Caro's acid : H_2SO_5

Marshal's acid ; $H_2S_2O_8$

S is + 6 in both

71)

NCERT pg # 269

72)

Solution :

$Ba(OH)_2 + 2HCl \rightarrow BaCl_2 + 2H_2O$

Eq of $Ba(OH)_2 = 0.01 \times 50 \times 2 /1000 = 1 \times 10^{-3}$ eq

Eq of HCl = $0.01 \times 25 /1000 = 2.5 \times 10^{-4}$ eq (limiting)

So neutralised eq = 2.5×10^{-4} eq

$\Delta H = 2.5 \times 10^{-4} \times 140$ kcal = 0.035 kcal = **35 cal**

Final answer : Option (2)

73)

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\square)$

$\Delta_C H(C_2H_4) = \Sigma(\Delta_f H)_P - \Sigma(\Delta_f H)_R$

74) $\Delta_r H = (2X + 435) - 2(565)$

$-430 = 435 + 2X - 1130$

X = 132.5 kJ/mol

75)

$n = \frac{5.8}{58} = 0.1$ mol

Now, for

2 mol → 5756 KJ

$0.1 \text{ mol} \rightarrow \frac{5756}{2} \times 0.1 = 287.8$ KJ

76) **Solution :**

Add the three given reactions:

$H_2O(g) + C(s) \rightarrow CO(g) + H_2(g) \quad (\Delta H = +131)$

$CO(g) + \frac{1}{2} O_2 \rightarrow CO_2(g)$ $(\Delta H = -282)$

$H_2(g) + \frac{1}{2}O_2 \rightarrow H_2O(g)$ $(\Delta H = 242)$

Summing cancels H_2O, CO, H_2 and yields $C + O_2 \rightarrow CO_2$. Total $\Delta H = 131 - 282 - 242 = -393$ kJ.

Final answer : option 1

77) $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightarrow NH_3$

$\Delta_f H = \frac{1}{2}BE_{N\equiv N} + \frac{3}{2}BE_{H-H} - 3BE_{N-H}$

$\Delta_f H = \frac{1}{2}x_1 + \frac{3}{2}x_2 - 3x_3$

78)

$Na_2SO_4(s) + H_2O(\square) \rightarrow Na_2SO_4(aq)$
excess

$\Delta H_1 = -2.34$ kJ mol^{-1} ..... (i)

$Na_2SO_4.10H_2O(s) + H_2O(\square) \rightarrow Na_2SO_4$ (aq),
excess

$\Delta H_2 = 78.87$ KJ mol^{-1} ..... (ii)

(i) - (ii)

$Na_2SO_4(s) + 10H_2O(\square) \rightarrow Na_2SO_4.10H_2O(s)$, $\Delta H_3 = ?$

$\Delta H_3 = \Delta H_1 - \Delta H_2 = -2.34$ kJ mol^{-1} -78.87 kJ mol^{-1}

$= -81.21$ kJ mol^{-1}

Final answer : option (1)

79)

Na(s) →Na(g) sublimation and Atomization

80)

P → $\Delta S_{sys} > 0$; $\Delta S_{surr} = 0$; $\Delta S_{univ} > 0$

$\boxed{P \rightarrow 1}$

Q → $\Delta S_{univ} = 0$; $\Delta S_{sys} > 0$; $\Delta S_{surr} < 0$

$\boxed{Q \rightarrow 1, 3, 4}$

R → $\Delta S_{univ} = 0$; $\Delta S_{sys} = 0$; $\Delta S_{surr} = 0$

$\boxed{R \rightarrow 3, 2}$

S → $\Delta S_{univ} = 0$; $\Delta S_{sys} > 0$; $\Delta S_{surr} < 0$

$\boxed{S \rightarrow 3, 1, 4}$

81) $(\Delta G)_{T,P} < 0$ for a spontaneous process

82) **Solution:**
Given: $\Delta H = -671.08$ kcal mol^{-1}
Reaction: $C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(l)$
$\Delta n_g = (2 - 3) = -1$
$\Delta H = -671.08$ kcal mol^{-1}
$\Delta E = \Delta H - \Delta ngRT$
At 27 °C = 300 K, R = 2 cal mol^{-1} K^{-1} = 0.002 kcal mol^{-1} K^{-1}
$\Delta n_gRT = (-1)(0.002 \times 300) = -0.6$ kcal
$\Delta E = -671.08 - (-0.6) = -670.48$ kcal mol^{-1}

Final Answer: Option (2)

83)

$W_{adi} = nC_V\Delta T$
$W_{adi} = 3 \times 27 \times 50$
$W_{adi} = 4050$ Cal

84)

Adiabatic, isothermal free expansion
$\Delta s = nR \ln \frac{V_2}{V_1}$

85)

$W = -P(V_2 - V_1)$
$W = -nR\ (T_2 - T_1)$
$W = -2 \times 2 \times 100$
$W = -400$ cal
Work done by the gas = 400 cal

86) **Solution:**
Condition: $\Delta G = \Delta H - T\Delta S < 0$
$\Delta H = 100$ kJ/mol = 100000 J/mol
$\Delta S = 400$ J/mol/K
$T > \Delta H / \Delta S$, $T > 250$ K
Final Answer: Option (2)

87)

Concept : This question is based on $\Delta S_{surr} = -\frac{\Delta H_{system}}{T}$.
Solution :
$\Delta S_{surr} = -\frac{2000\ kJ/mol}{400\ K}$
$\Delta S_{surr} = -5$ kJ/mol. K

Final answer : 2

88) $\Delta H = \Delta U + P\Delta V$

$= -2 + (1 \times 10) \times \frac{100}{1000}$

$= -1$ kJ

89) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$y = c + mx$

slope $= -\Delta S^\circ = (-1)$

$\Delta G^\circ = -500 - 300 \times (1)$

$\Delta G^\circ = -500 - 300$

$\Delta G^\circ = -800$ cal/mol

90)

Theory based on Le-chatelier's principle.

BIOLOGY

91)

NCERT PG. 5

92) NCERT, Pg # 30, Para.-3.0

93) NCERT Pg.# 7

94) NCERT XI Pg # 23

95) NCERT (XI) Pg. # 128, 129

96) NCERT (XI) Pg # 118

97) NCERT Pg. # 11, 19

98) NCERT XI Page No. # 20, 21

99) NCERT XI(E)/(H) Page No. # 20, 27

100) NCERT (XIth) Pg. # 26, 2nd para

101)

NCERT XI, # 20, 21

102) NCERT (XI) Pg. # 19
Bacteria are sole members of monera not archaebacteria

103) NCERT (XI) Pg. # 23,24

104) NCERT (XIth) Pg. # 32, 33, 34

105)

NCERT (XIth) 13, 26

106)

NCERT-XII, Pg. # 31

107) NCERT-(XIth) Pg. # 24

108) NCERT Pg. No. # 13, 27

109) NCERT Page No. 38

110)

NCERT-(XIth) Pg. # 30, 32

111) NCERT-XII, Pg. # 39

112) NCERT Pg. # 35

113)

NCERT-(XIth) Pg. # 33

114) NCERT (XIth) Pg. # 28

115)

NCERT (XIth) Pg. # 25

116) NCERT-XI, Pg. # 33, Fig.-3.4 (b)

117) NCERT Pg. # 12

118) NCERT-XI Pg. # 9

119)

NCERT (XIth) Pg. # 7

120)

NCERT XI pg.# 9

121) NCERT-XI, Para-1.3.7

122)

NCERT-(XI) Pg.6, 7, 8

123)

NCERT-(XI) Pg.# 13

124)

NCERT-(XI) Pg.# 15,17,18

125)

NCERT-(XIth) Pg. # 20

126)

NCERT-(XIth) Pg.# 15

127) NCERT Pg.#15,16.

128) NCERT XI 23

129) Module

130) NCERT-XI Pg.#11
Hint: C, D & E correct.

131) NCERT-XI, Pg. # 8

132) NCERT Pg.#29-30

133) NCERT Pg. # 30, 32, 33

134)

NCERT XI Pg.#30,32,33,34

135) NCERT XI Pg 30

136)

NCERT (XI) Pg. # 50

137)

NCERT (XI) Pg. # 49

138) NCERT (XI) Pg. # 51,59

139) NCERT Pg. # 61

140) NCERT Pg. # 42

141)

NCERT (XI) Pg. # 56

142) NCERT-XI, Pg. # 46-51

143) NCERT Pg. # 57

144) NCERT (XI) Pg. # 56

145) NCERT Pg. # 47

146)

NCERT, Pg. # 39

147) NCERT Pg.# 40, 41, 42

148) NCERT, Pg. # 41

149)

NCERT Pg. # 54

150)

NCERT Pg. # 54

151) NCERT Pg. # 55

152)

NCERT-XI, Pg. # 57

153) NCERT, Pg # 46, 48

154) NCERT Pg.# 41, 42, 44, 45

155) NCERT XIth Page # 57

156)

NCERT XIIth, Pg.No.102 Q-2 7.1.1

157)

NCERT XIth Pg.No.103 Para No.2

158) NCERT XI, Pg. No :- 101, 102

159) NCERT XI, Pg. No :- 102

160) NCERT XI, Pg. No :- 101

161) NCERT Page No. 197

162) NCERT Pg. # 103

163)

NCERT Pg. # 101 (figure-7.1)

164) NCERT Pg. # 113

165) NCERT Pg.#114 (E), 114 (H)

166) NCERT Pg.#114 (E), 114 (H)

167)

NCERT Pg. # 112

168)

NCERT Pg. # 113

169)

NCERT Pg. # 114

170) Module.

171) Module.

172)

NCERT Pg. # 119

173)

NCERT Pg # 118

174) NCERT-XII Page No. 117

175) NCERT Page No. 117

176) NCERT Page No. 118

177) NCERT Page - 118 and 119

178) NCERT Pg # 83

179) NCERT Pg. # 118

180) NCERT Page - 116 to 119